

# NEM 2026 – Getting the most out of TwinCAT Scope

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## NEM 2026 : Agenda

- Introduction, Licensing & Messaging
- Scope Basics
- Charts, Sampling & Triggers
- Layer Charts
- Dynamic Styles
- Vision Integration
- Reporting and Data Export Formats
- Additional Capabilities
- Tips & Tricks
- Q & A



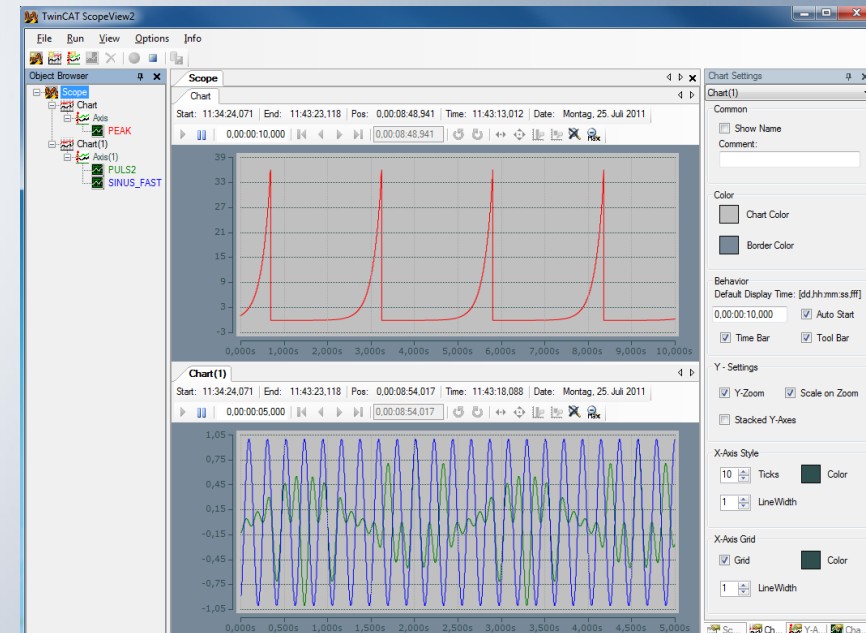
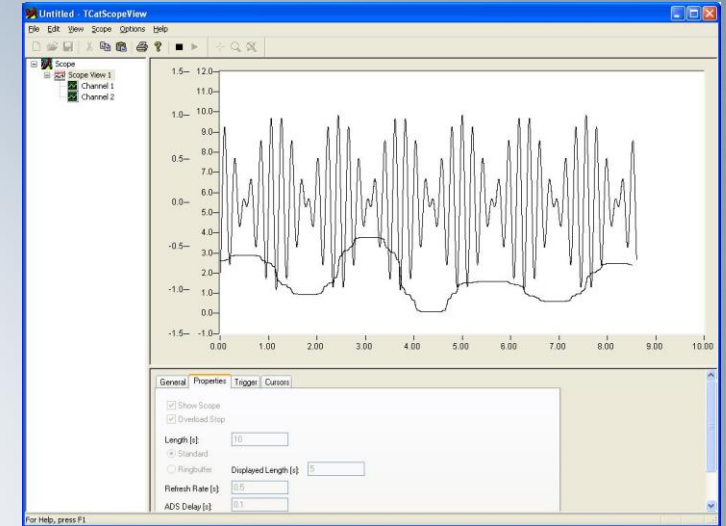
## NEM 2026 : Agenda

- **Introduction, Licensing & Messaging**
- Scope Basics
- Charts, Sampling & Triggers
- Chart Configuration & Visualization
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## Introduction - History

- **TwinCAT ScopeView 1** was introduced in TwinCAT 2
  - Royalty free version
- **TwinCAT ScopeView 2** was later introduced (TS3300)
  - License required (30 day demo available)
  - Viewer / Server
  - Long time recording
  - Auto Scaling
- **TwinCAT 3 Scope**
  - Scope 2 + Visual Studio & TwinCAT 3
  - Functionality based on licensing level





## Introduction - Licensing

### TwinCAT 3 Scope Base

- Integrated into TwinCAT Engineering (TE1000) Setup
- View and local Server
- Machine start-up
- Free of charge

### TwinCAT 3 Scope View Professional

- TE1300 with extended functions:
  - long term records, Subsave trigger, auto restart, auto save,
  - extended export formats,
  - vision integration and
  - .NET Control Integration
- Process control
- Subject to license terms

### TwinCAT 3 Scope Server

- Remote Server – necessary for automatic Server records without Scope View
  - ⇨ FB\_ScopeServerControl in TcUtilities library or Headless Mode
- Subject to license terms

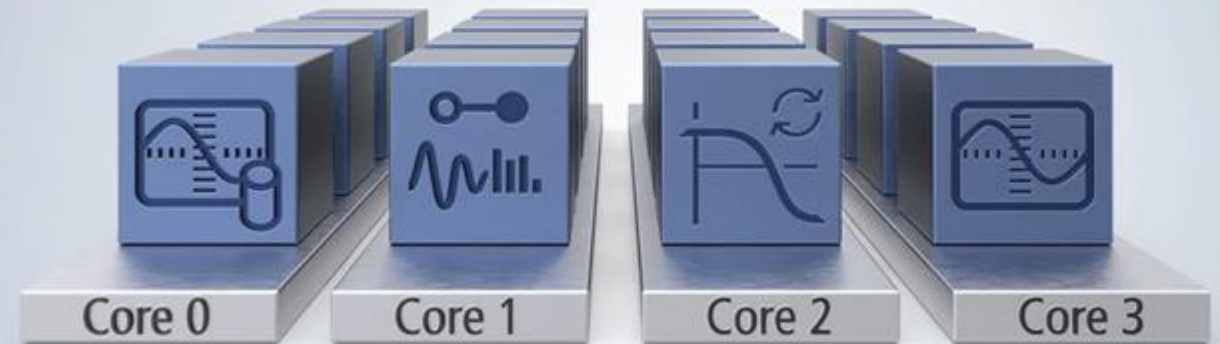
Overview of functionalities provided by license:

| Features                          | Scope Base |      | Scope Server Full Licence |                      | Scope View Professional |                      |
|-----------------------------------|------------|------|---------------------------|----------------------|-------------------------|----------------------|
|                                   | Server     | View | Server                    |                      | View                    |                      |
|                                   |            |      | Full                      | 7 days trial version | Full                    | 7 days trial version |
| <b>General:</b>                   |            |      |                           |                      |                         |                      |
| Free of charge                    | ✓          | ✓    | ✗                         | ✓                    | ✗                       | ✓                    |
| Local record                      | ✓          | ✓    | ✓                         | ✓                    | ✓                       | ✓                    |
| Remote Record using Target Server | ✗          | ✓    | ✓                         | ✓                    | ✓                       | ✓                    |
| Remote Record using Local Server  | ✓          | ✓    | ✓                         | ✓                    | ✓                       | ✓                    |
| Scope Control Integration         | -          | ✗    | -                         | -                    | ✓                       | ✓                    |
| Long Time Records > 1h            | -          | ✗    | -                         | -                    | ✓                       | ✗                    |
| Ring Buffer                       | ✓          | ✓    | ✓                         | ✓                    | ✓                       | ✓                    |
| Auto Save                         | -          | ✗    | -                         | -                    | ✓                       | ✓                    |
| Auto Export                       | -          | ✗    | -                         | -                    | ✓                       | ✗                    |
| Restart Record                    | -          | ✗    | -                         | -                    | ✓                       | ✓                    |
| View Multicore Support            | -          | ✓    | -                         | -                    | ✓                       | ✓                    |

[https://infosys.beckhoff.com/content/1033/te13xx\\_tc3\\_scopeview/182261131.html?id=2470127920446856668](https://infosys.beckhoff.com/content/1033/te13xx_tc3_scopeview/182261131.html?id=2470127920446856668)

## TwinCAT 3 Scope

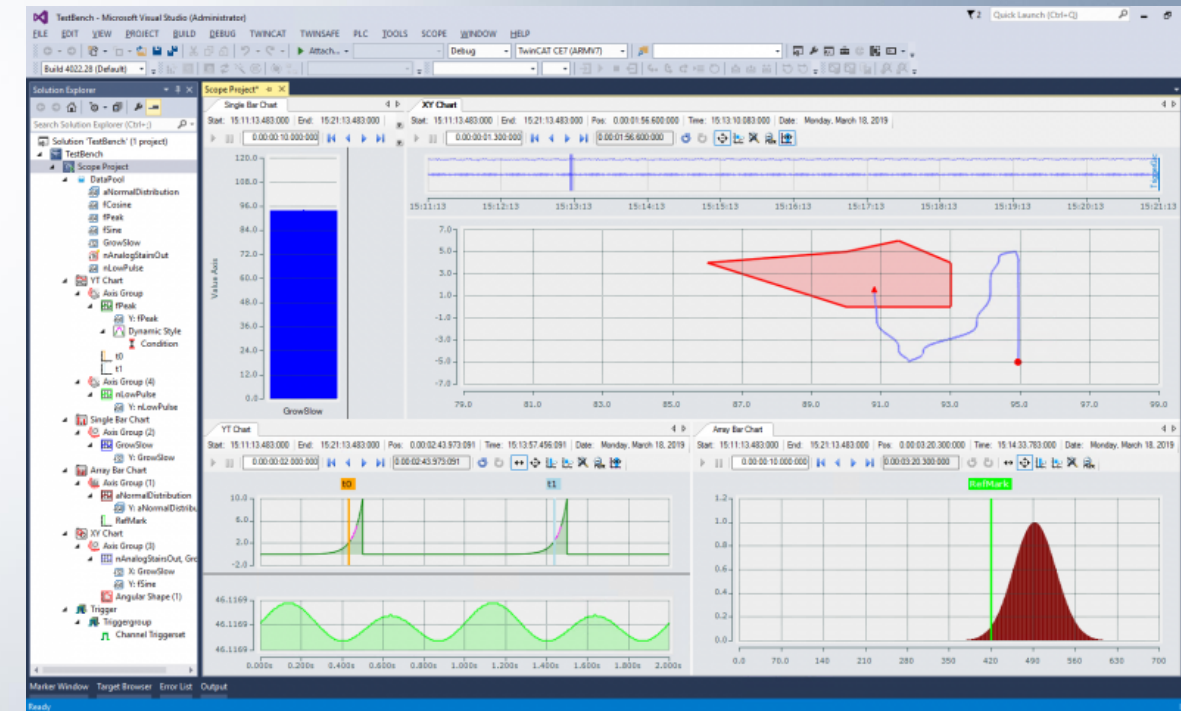
- A high-performance, real-time data acquisition and visualization tool
- Enables engineers to record, correlate, and analyze machine signals
- Critical for commissioning, troubleshooting, and optimization.



TwinCAT 3 Scope: Multi-core Oscilloscope for Big Data

# TwinCAT Scope Capabilities

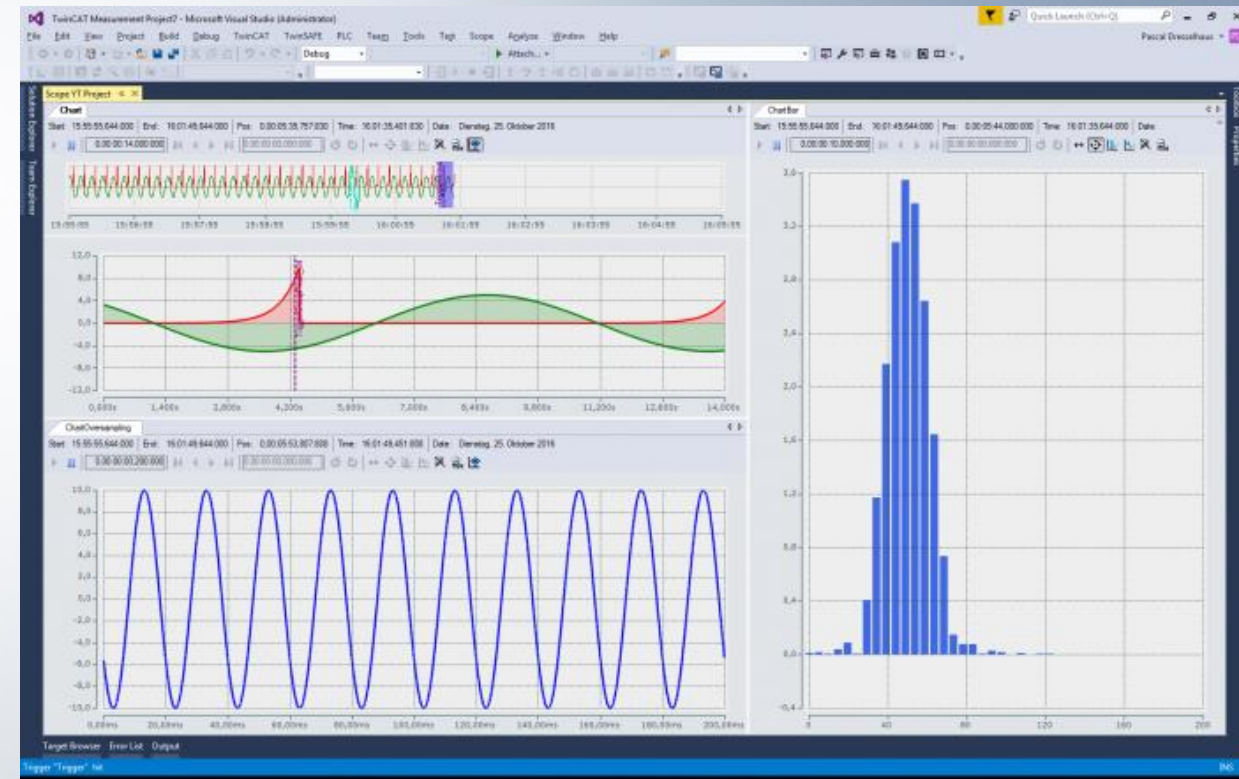
- Real-time recording of real time data
- High-resolution sampling with EtherCAT synchronization
- Trigger-based capture (pre/post event recording)
- Multi-channel visualization (time charts, XY charts)
- Layered charting for signal correlation
- Dynamic styles for condition-based highlighting
- Integration with vision data via **TwinCAT Vision**
- Data export and reporting





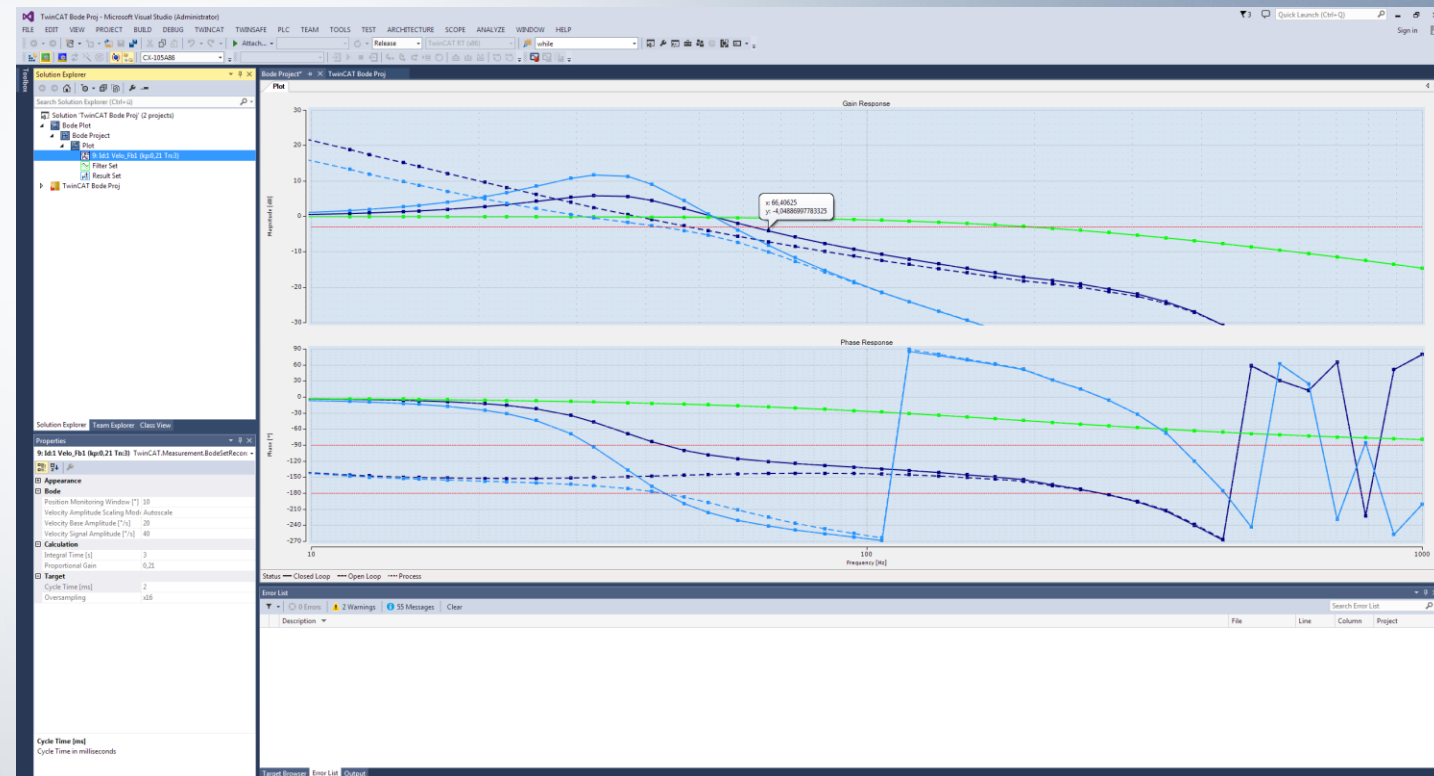
## TwinCAT Scope Benefits

- Enable faster commissioning and more consistent support workflows
- Provide clear documentation and proof of system behavior
- Reduce troubleshooting effort
- Gain visibility into real-time machine behavior
- Improve machine performance and reliability
- Replace guesswork with data-driven analysis



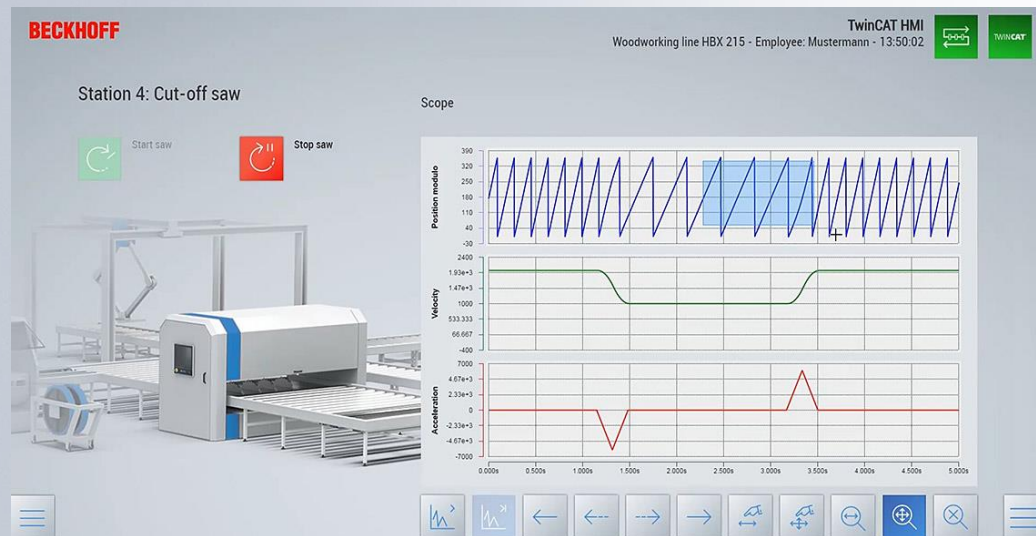
## TwinCAT 3 Scope Common Use Cases

- Machine commissioning and signal validation
- Motion tuning (position, velocity, following error)
- Debugging I/O and sequencing issues
- Cycle time and performance analysis
- Capturing intermittent faults



## TwinCAT 3 Scope Common Use Cases

- Correlating machine states with motion and I/O using Layer Charts
- Highlighting anomalies automatically using Dynamic Styles
- Synchronizing and analyzing vision inspection results with machine events
- High-speed event capture using oversampling and precise triggers
- Creating reusable diagnostic templates for standard machines
- Generating reports for internal teams or end customers



### Quality management

BECKHOFF

Inspection according to manufacturer's specification

2022-05-02 12:17:08

#### General test information

|               |                                   |
|---------------|-----------------------------------|
| Manufacturer  | Beckhoff Automation GmbH & Co. KG |
| Test bench    | D2-750 bench 4                    |
| Description   | Drilling Machine MS 251           |
| Type          | MS 251 C-BE, RM3                  |
| Serial number | 253-5553-9421                     |

#### Analytics Workbench Overview

##### Additional Information

Operator: Max Mustermann

##### Measurement Information

Analysis Start Time: 02.05.2022 12:05:41

Analysis End Time: 02.05.2022 12:05:56

#### Rotations and Vibrations



## NEM 2026 : Agenda

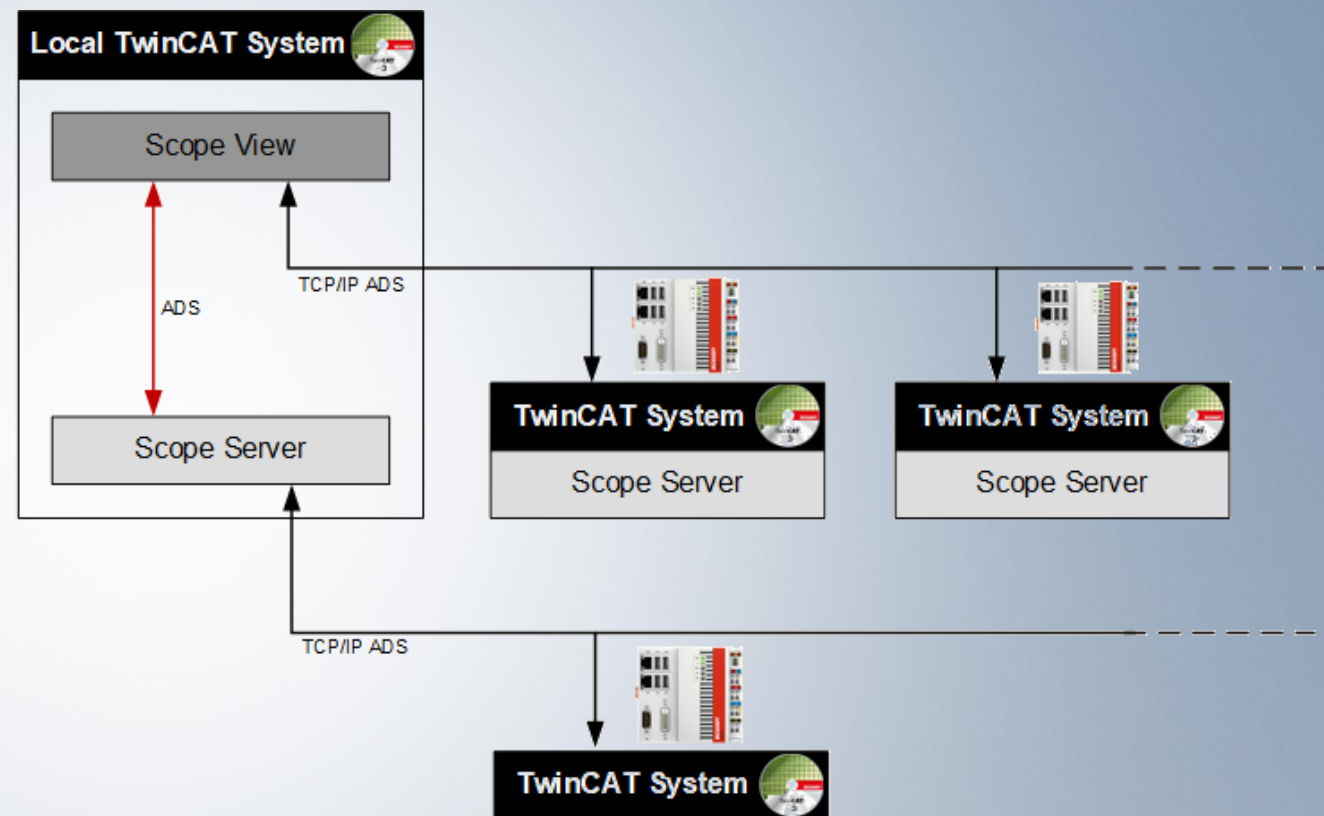
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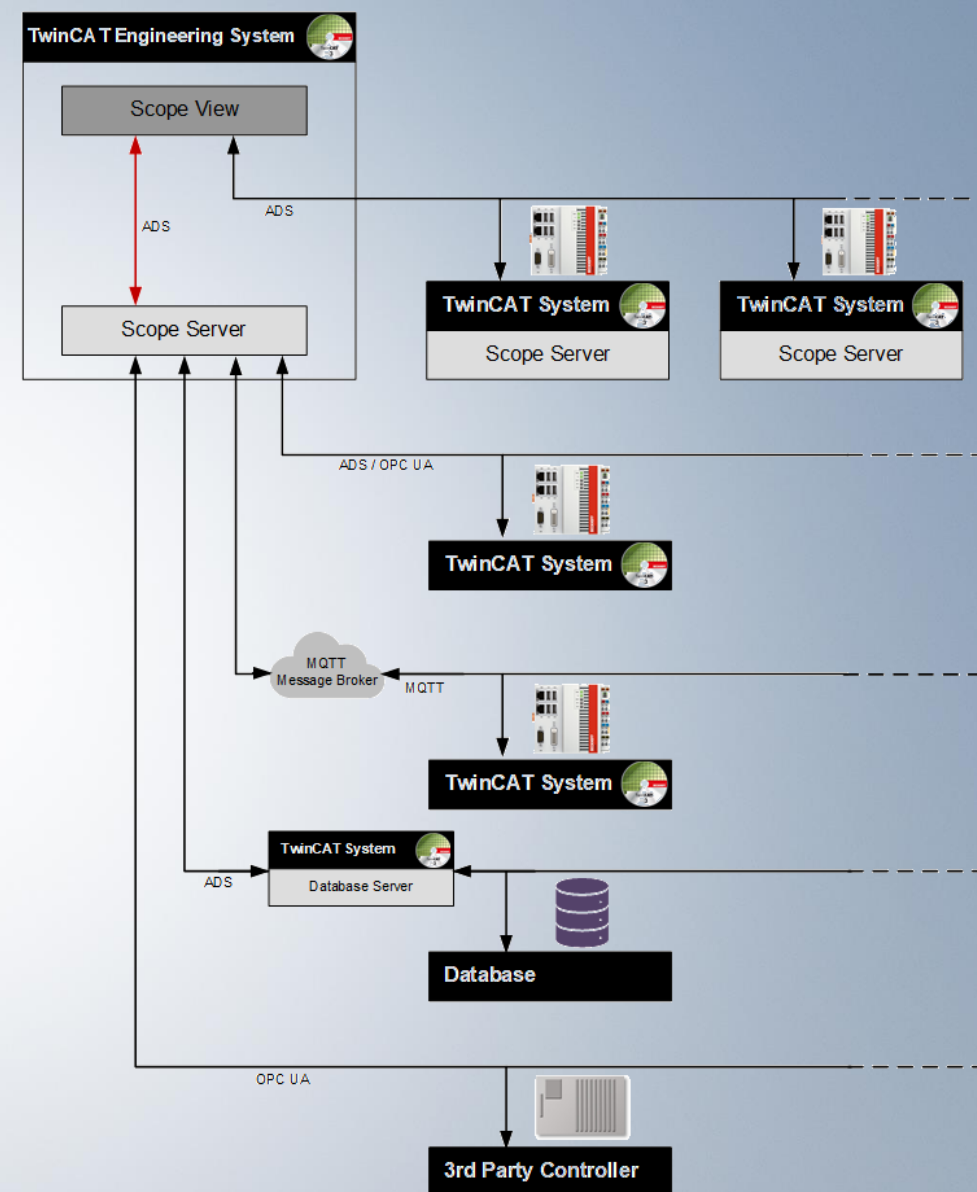
## Basics – Architecture

- Client / Server model
  - Scope **View** is the client
  - Scope **Server** is the backend



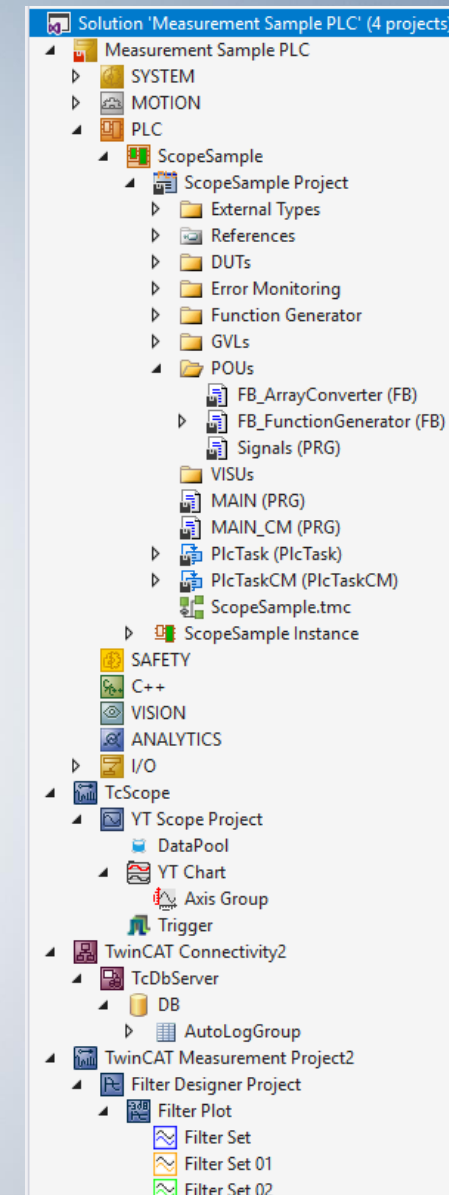
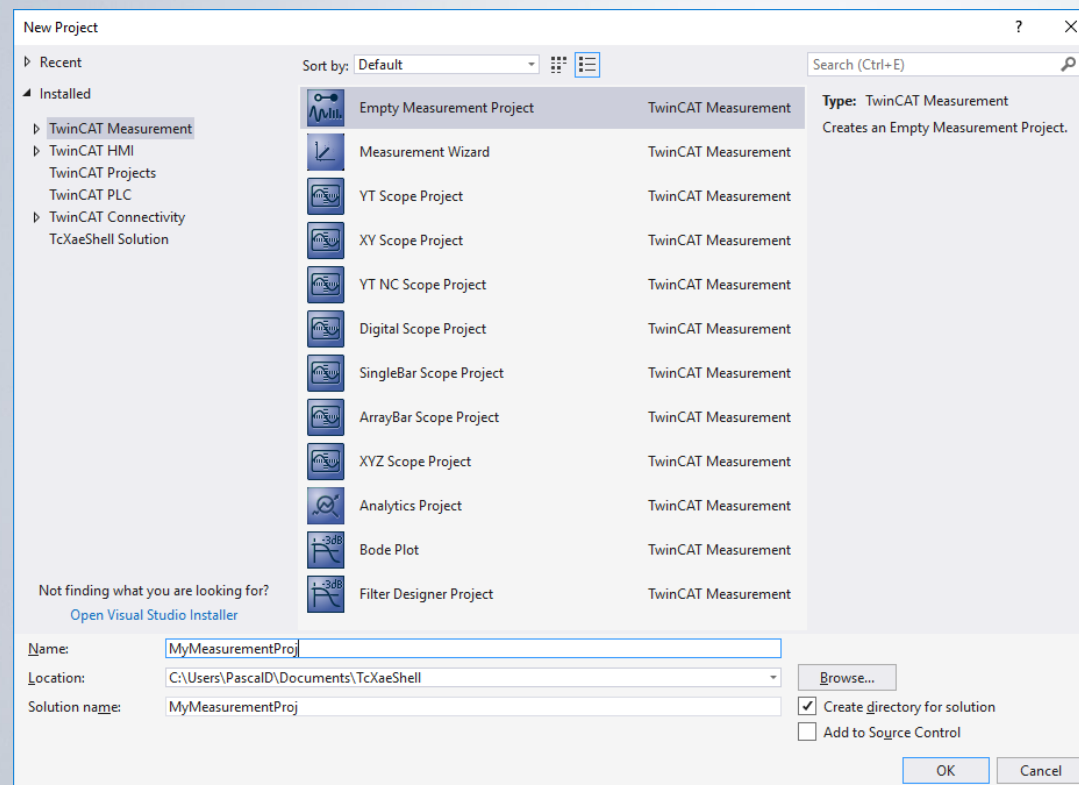
# Basics – Communication

- Extended communication (beyond ADS)
  - OPC UA
  - Analytics
    - MQTT Stream
    - Logger Files
  - Scope acquisitions (svdx)
  - TwinCAT Database Server



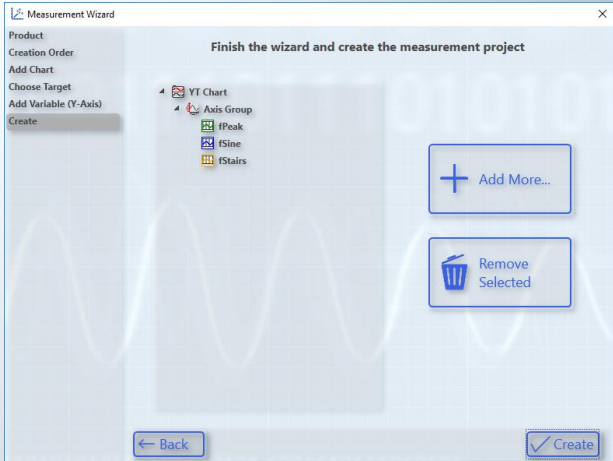
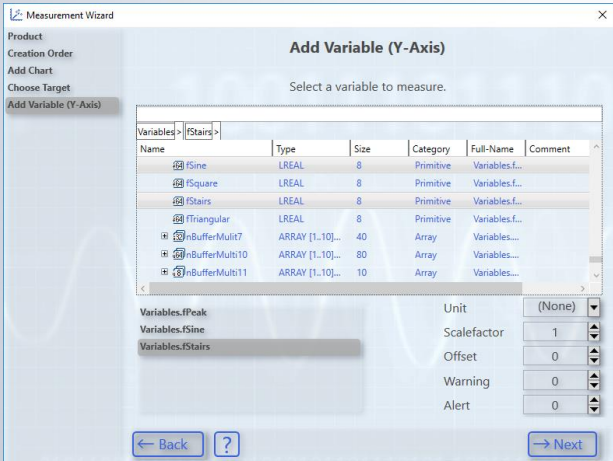
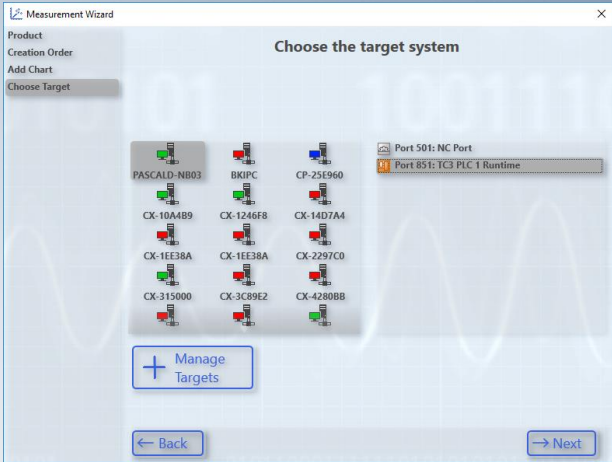
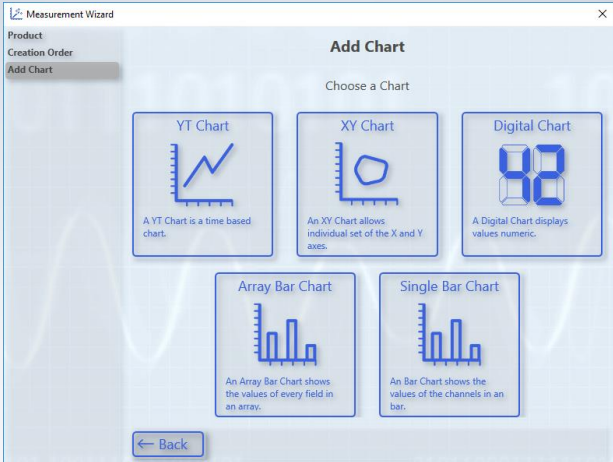
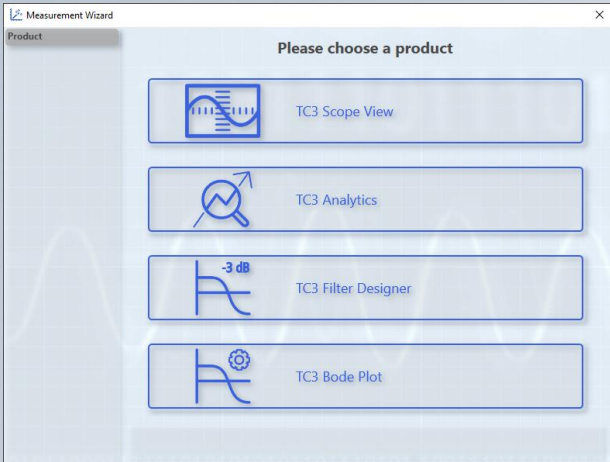
# Base configuration: Measurement Project

- Integration as an individual project
- Can exist in parallel to TwinCAT in one solution
- Creates synergies between the TC-products



# Base configuration: Measurement Project Wizard

For beginners: The Measurement Wizard





## Base configuration: Target Browser

- Easy selection of channels with the Target Browser
- Integrated into Visual Studio as an individual Tool-Window
- Multi-select and Drag & Drop are possible
- Filter settings for variables

Target Browser

ADS TcAnalytics TcAnalytics File OpcUa

Enter Filter...

Variables > fPeak >

| Name                | Type             | Size  | Category  | Full-Name                     | Comment | Subitems | Unit | Context-Mask | Index-Group | Index-Offset | Attributes (Instance) | Attributes (Type) |
|---------------------|------------------|-------|-----------|-------------------------------|---------|----------|------|--------------|-------------|--------------|-----------------------|-------------------|
| aBufferSine         | ARRAY [1..10]... | 80    | Array     | Variables.aBufferSine         |         | 10       |      | 0            | 4040        | 6DE38        | { Area : Local }      | none              |
| aBufferSquare       | ARRAY [1..10]... | 80    | Array     | Variables.aBufferSquare       |         | 10       |      | 0            | 4040        | 6DE88        | { Area : Local }      | none              |
| aBufferStairs       | ARRAY [1..10]... | 80    | Array     | Variables.aBufferStairs       |         | 10       |      | 0            | 4040        | 6DD98        | { Area : Local }      | none              |
| aBufferTriangular   | ARRAY [1..10]... | 80    | Array     | Variables.aBufferTriangular   |         | 10       |      | 0            | 4040        | 6DED8        | { Area : Local }      | none              |
| aNormalDistribution | ARRAY...         | 16000 | Array     | Variables.aNormalDistribution |         | 2000     |      | 0            | 4040        | 6E1B0        | { Area : Local }      | none              |
| aSinusTerminal      | ARRAY [1..10]... | 10    | Array     | Variables.aSinusTerminal      |         | 10       |      | 0            | F020        | 6E178        | { Area : Input }      | none              |
| bAnalogOut          | BOOL             | 1     | Primitive | Variables.bAnalogOut          |         | 0        |      | 0            | F030        | 6C3A5        | { Area : Output }     | none              |
| fAM                 | LREAL            | 8     | Primitive | Variables.fAM                 |         | 0        |      | 0            | 4040        | 6DC68        | { Area : Local }      | none              |
| fPeak               | LREAL            | 8     | Primitive | Variables.fPeak               |         | 0        |      | 0            | 4040        | 6DC70        | { Area : Local }      | none              |
| fPulse              | LREAL            | 8     | Primitive | Variables.fPulse              |         | 0        |      | 0            | 4040        | 6DC78        | { Area : Local }      | none              |
| fSawtooth           | LREAL            | 8     | Primitive | Variables.fSawtooth           |         | 0        |      | 0            | 4040        | 6DC80        | { Area : Local }      | none              |
| fSine               | LREAL            | 8     | Primitive | Variables.fSine               |         | 0        |      | 0            | 4040        | 6DC88        | { Area : Local }      | none              |
| fSquare             | LREAL            | 8     | Primitive | Variables.fSquare             |         | 0        |      | 0            | 4040        | 6DC90        | { Area : Local }      | none              |
| fStairs             | LREAL            | 8     | Primitive | Variables.fStairs             |         | 0        |      | 0            | 4040        | 6DC98        | { Area : Local }      | none              |
| fTriangular         | LREAL            | 8     | Primitive | Variables.fTriangular         |         | 0        |      | 0            | 4040        | 6DCA0        | { Area : Local }      | none              |
| nAnalogSine         | ARRAY [1..10]... | 20    | Array     | Variables.nAnalogSine         |         | 10       |      | 0            | F020        | 6E182        | { Area : Input }      | none              |
| nAnalogSineOut      | INT              | 2     | Primitive | Variables.nAnalogSineOut      |         | 0        |      | 0            | F030        | 6DBE6        | { Area : Output }     | none              |

Target Browser Error List Output

# Base configuration: PLC Control

## Selection of variables from TwinCAT PLC Control

The screenshot displays the TwinCAT 3 Scope interface. On the left, the 'Signals (PRG)' tree is expanded, showing the project structure. The 'TcScope' folder is selected, and the 'YT Scope Project' is expanded, showing the 'DataPool' and 'YT Chart' components. The 'YT Chart' component is selected, and the 'Axis Group' is expanded, showing the 'Trigger' component. The 'Trigger' component is selected, and the 'TwinCAT Connectivity2' folder is expanded, showing the 'TcDbServer' component. The 'TcDbServer' component is selected, and the 'DB' folder is expanded, showing the 'AutoLogGroup' component. The 'AutoLogGroup' component is selected, and the 'TwinCAT Measurement Project2' folder is expanded, showing the 'Filter Designer Project' component. The 'Filter Designer Project' component is selected, and the 'Filter Plot' component is expanded, showing the 'Filter Set' component. The 'Filter Set' component is selected, and the 'Filter Set 01' component is expanded, showing the 'Filter Set 02' component. The 'Filter Set 02' component is selected.

On the right, the PLC control code is displayed. A context menu is open over the code, showing the following options:

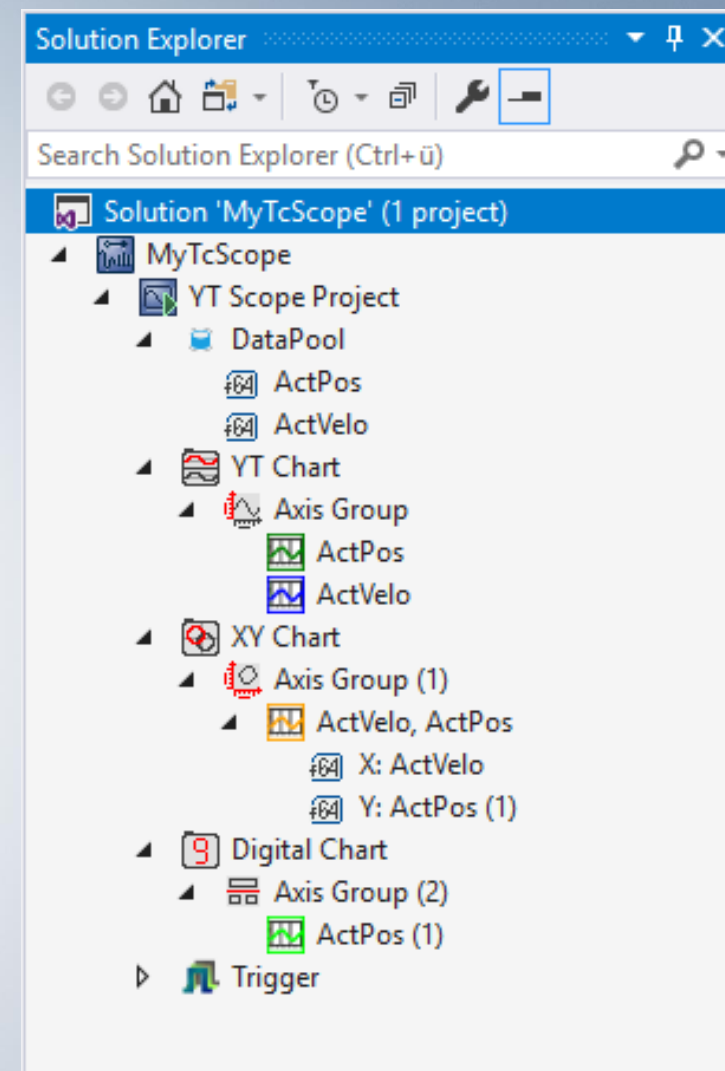
- Add Watch
- Add to scope
- Browse Call Tree
- Go To Definition F12
- Go To Instance
- Find All References Shift+F12
- Input Assistant... F2
- Auto Declare...
- Refactoring
- Toggle Breakpoint F9
- Display Mode
- Copy Ctrl+C
- Delete Del
- Select All Ctrl+A

The code in the background includes the following lines:

```
16 nDutyCycle := ,
17 bNoise FALSE := bNoisePulse FALSE,
18 aSignal => aBufferPulse );
19 fPulse 0 := aBufferPulse[1] 0 ;
20
21 //Peak
22 fbGeneratorPeak(bEnable TRUE := bEnablePeak TRUE,
23 fFrequency 0.1 := fSetFrequencyPeak 0.1,
24 fAmplitude 10 := fSetAmplitudePeak 10,
25 fOffset := ,
26 nDutyCycle := ,
27 bNoise FALSE := bNoisePeak FALSE,
28 aSignal => aBufferPeak );
29
30
31 //S
32 fb0
33
34
35
36
37
38
39 fSt
40
41 //S
42 fb0
43
44
45
46
47
48
49 fSawtooth 4.29 := aBufferSawtooth[1] 4.29 ;
50
```

## Base configuration: Data Pool

- DataPool: data that is *actually* read
  - Regardless of chart configs
  - Single acquisition for multiple functions
  - Holds data source, connection, sampling rate, etc. config
- Record first, render later



## NEM 2026 : Agenda

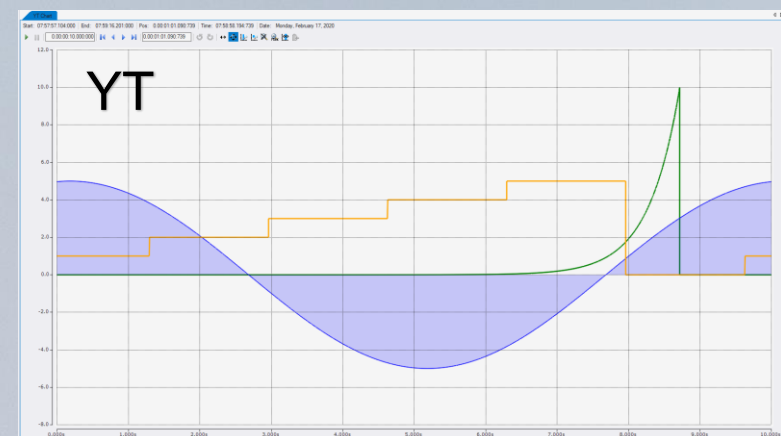
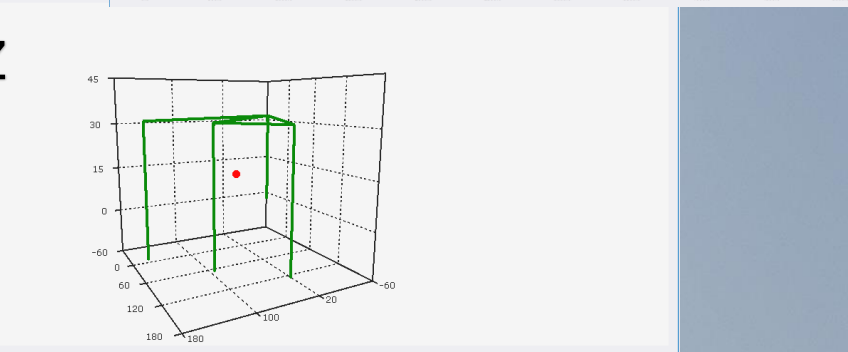
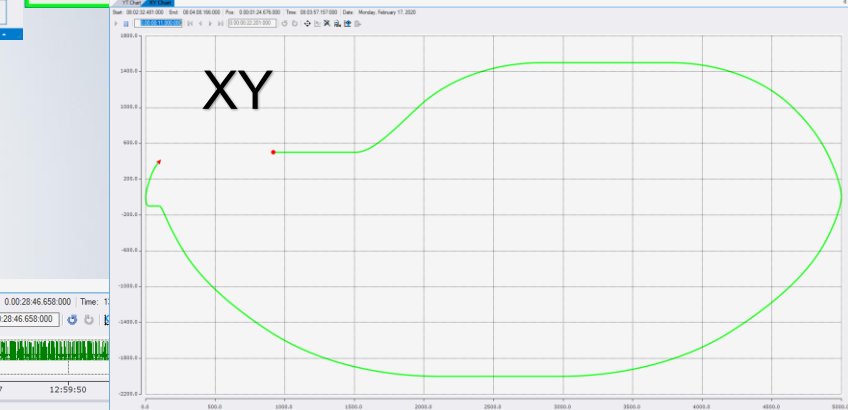
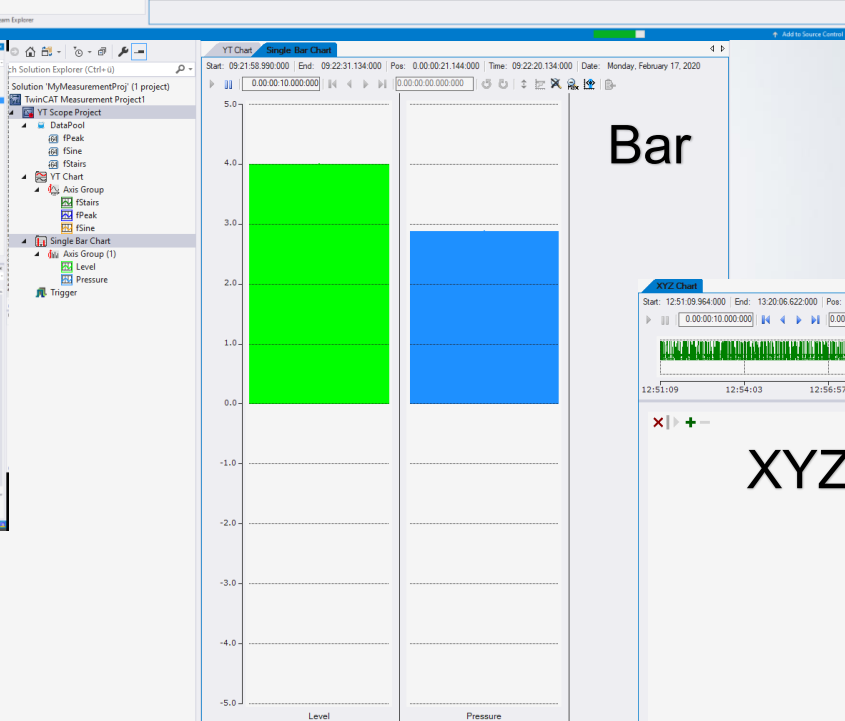
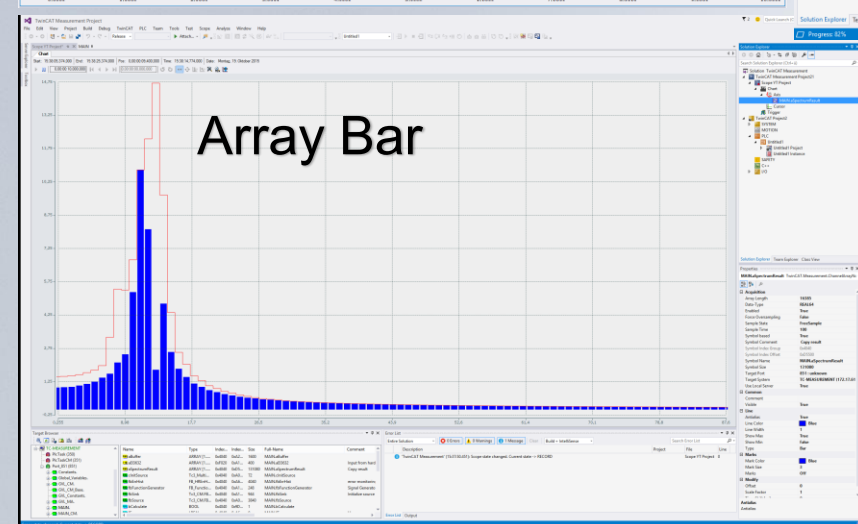
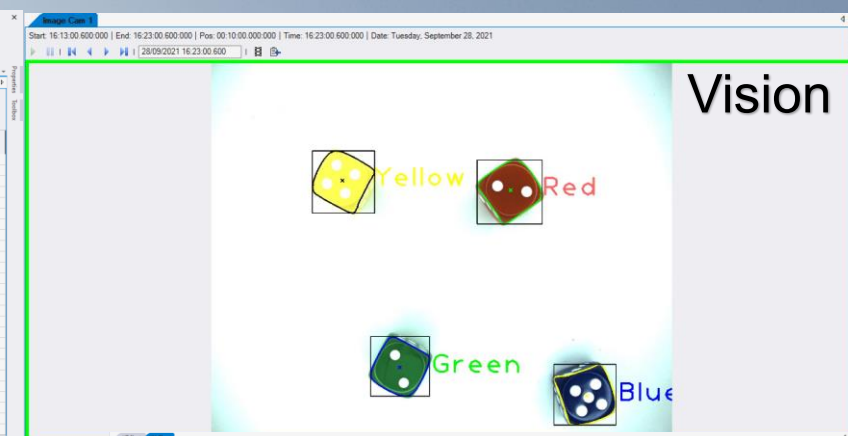
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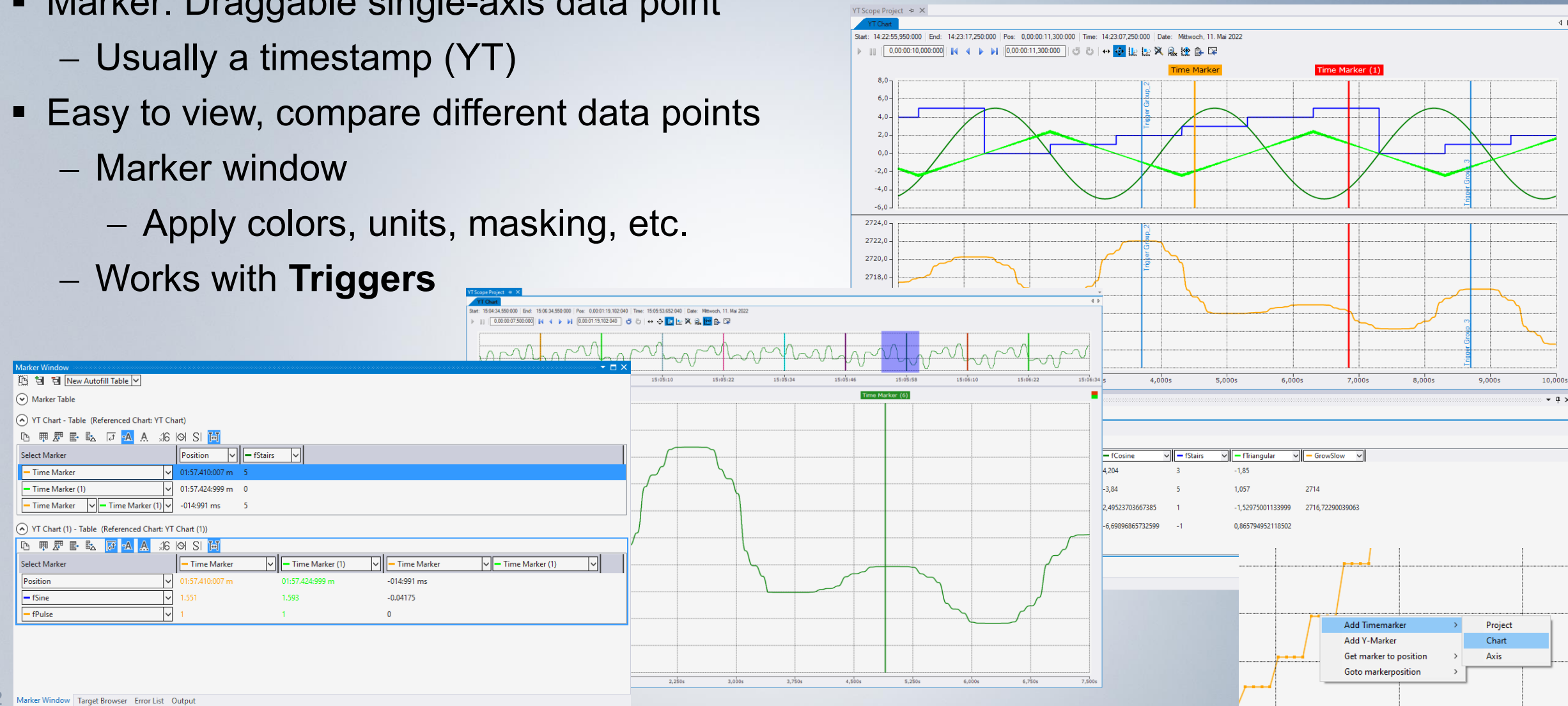
## Charting: Chart Types

# BECKHOFF

[illegible]

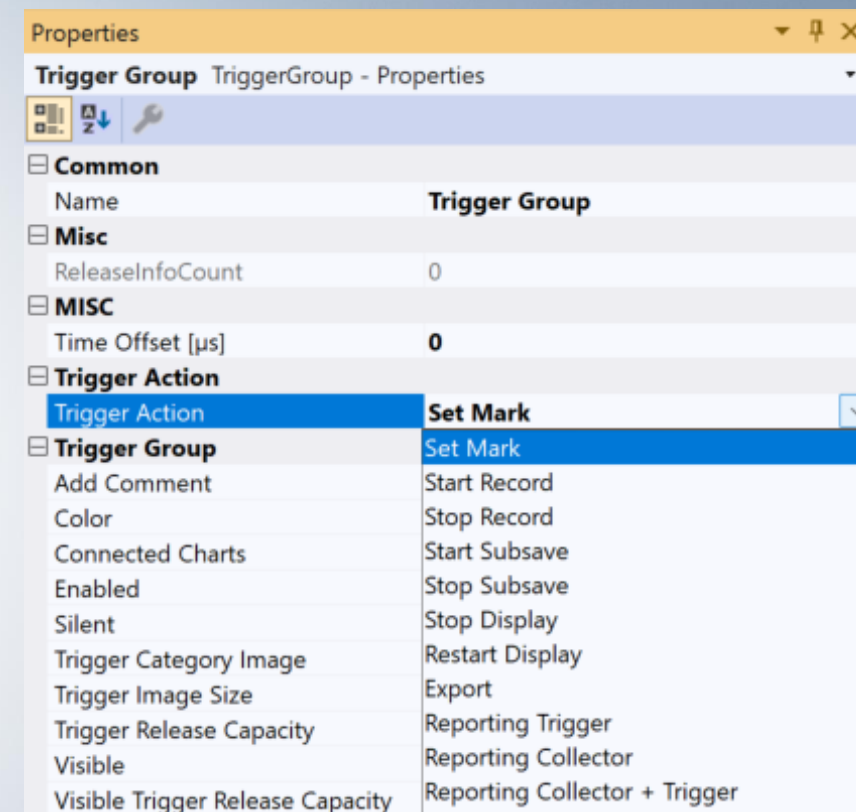
A digital multimeter display showing a voltage reading of 4.363V. The display is split into two sections: 'Voltage' on the left and 'Digital' on the right. The digits are blue on a grey background. Below the digits are range markers for -5 and 5.

- Marker: Draggable single-axis data point
  - Usually a timestamp (YT)
- Easy to view, compare different data points
  - Marker window
    - Apply colors, units, masking, etc.
  - Works with **Triggers**



# Triggers

- Trigger: Automate actions in Scope
- Actions
  - **Set Mark**
  - **Start/Stop Record**
  - Stop/Stop Subsave
  - Restart Display
  - Export
  - Reporting Trigger, Collector, Collector + Trigger
  - Trigger Group properties change based on Trigger Action Type







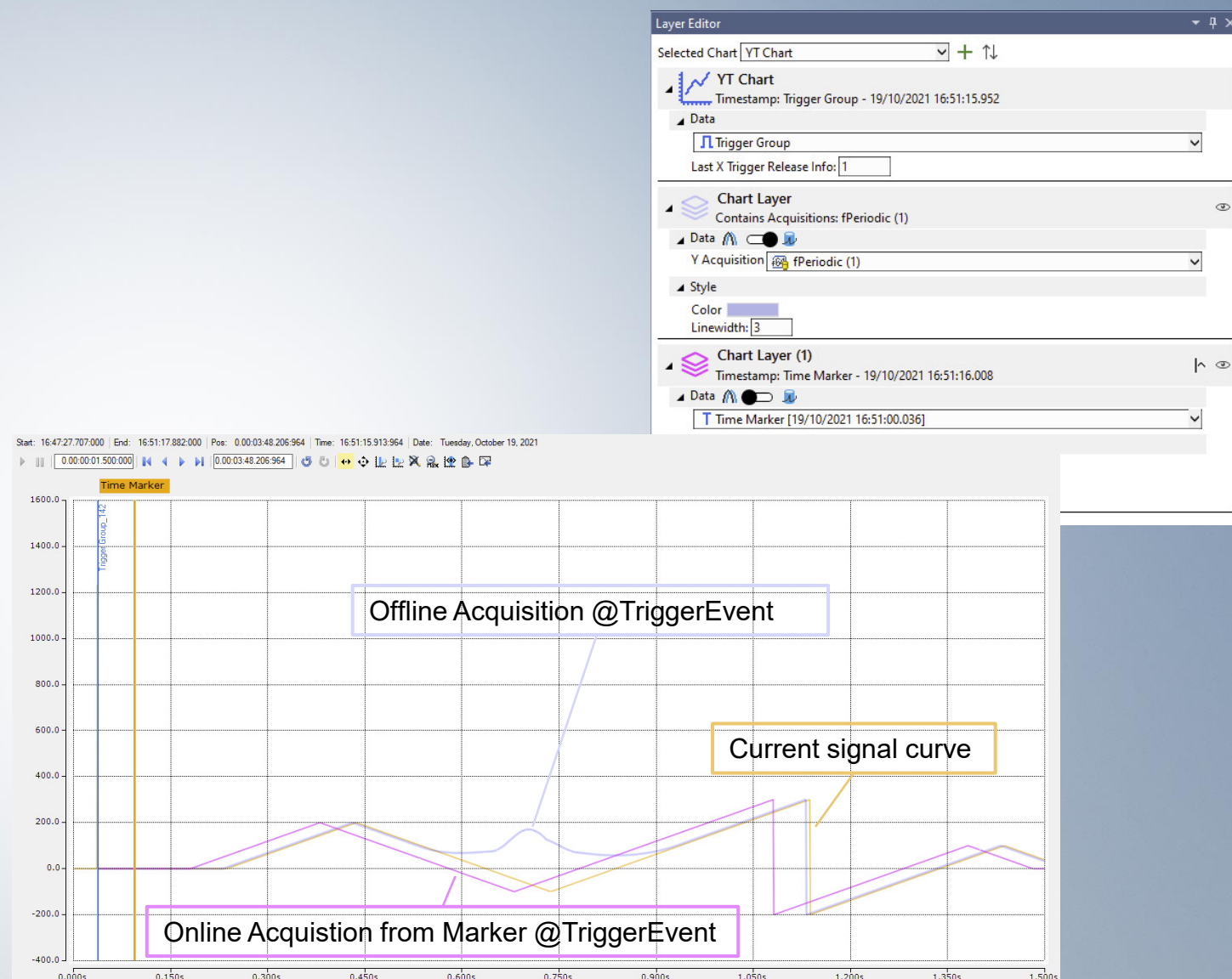
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## TwinCAT Scope Layer Chart

- Used to compare signal curves
  - Online or offline in YT or XY
- Can compare channel *with itself*
  - Via time offsets
  - Or with offline acquisitions



# Lab 1 – The Basics

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Clone repo:

[https://github.com/bstiffler582/2026\\_NEM\\_Scope](https://github.com/bstiffler582/2026_NEM_Scope)

Follow Lab 1 in **guide.md** file  
(best viewed in browser)



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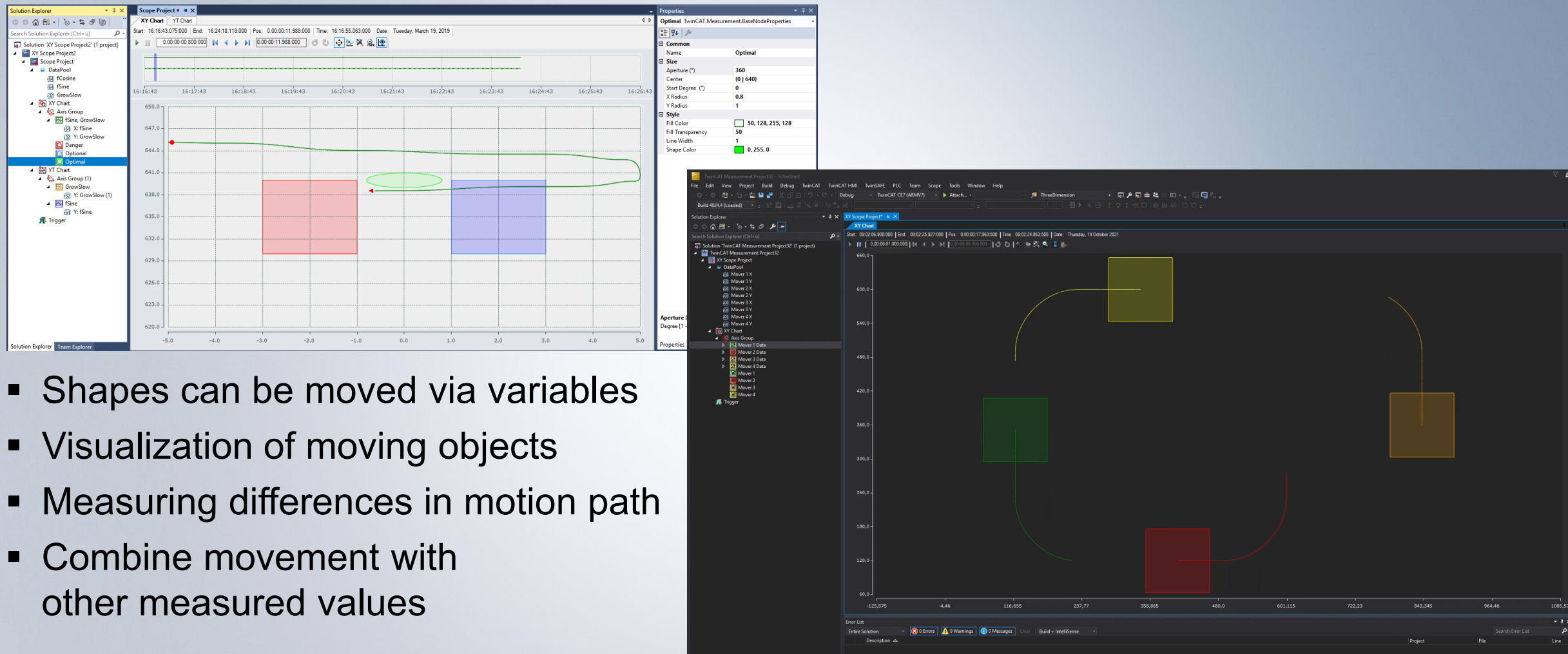


## Dynamic Style

- Dynamic Style – change style by variable from configuration
  - Visible/Invisible
  - Line color
  - Line style (dotted...)
  - Line width
- Multiple conditions
- Useful for...
  - NC, CNC, XTS
  - Classification like threshold monitoring
  - Show deepness



- Shapes for XY Charts with moving function



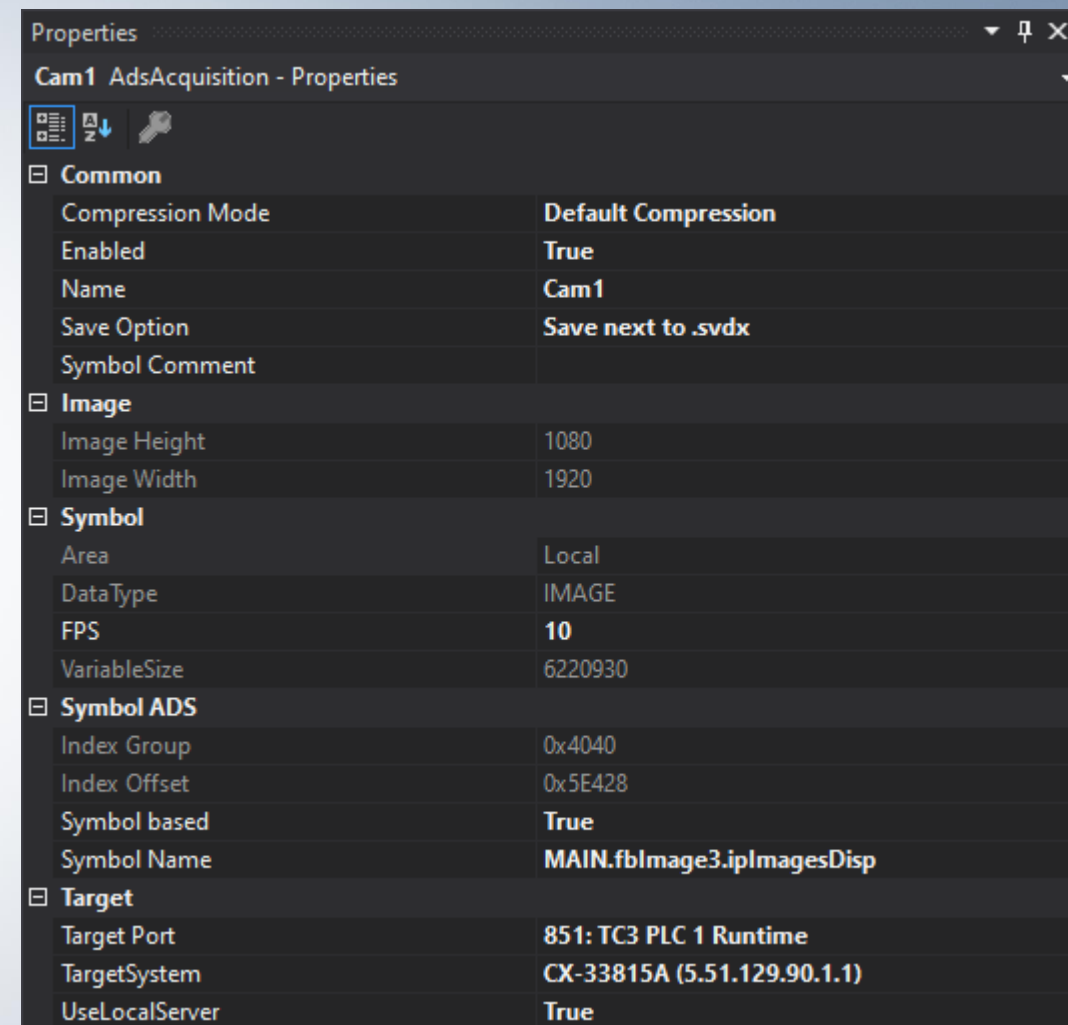
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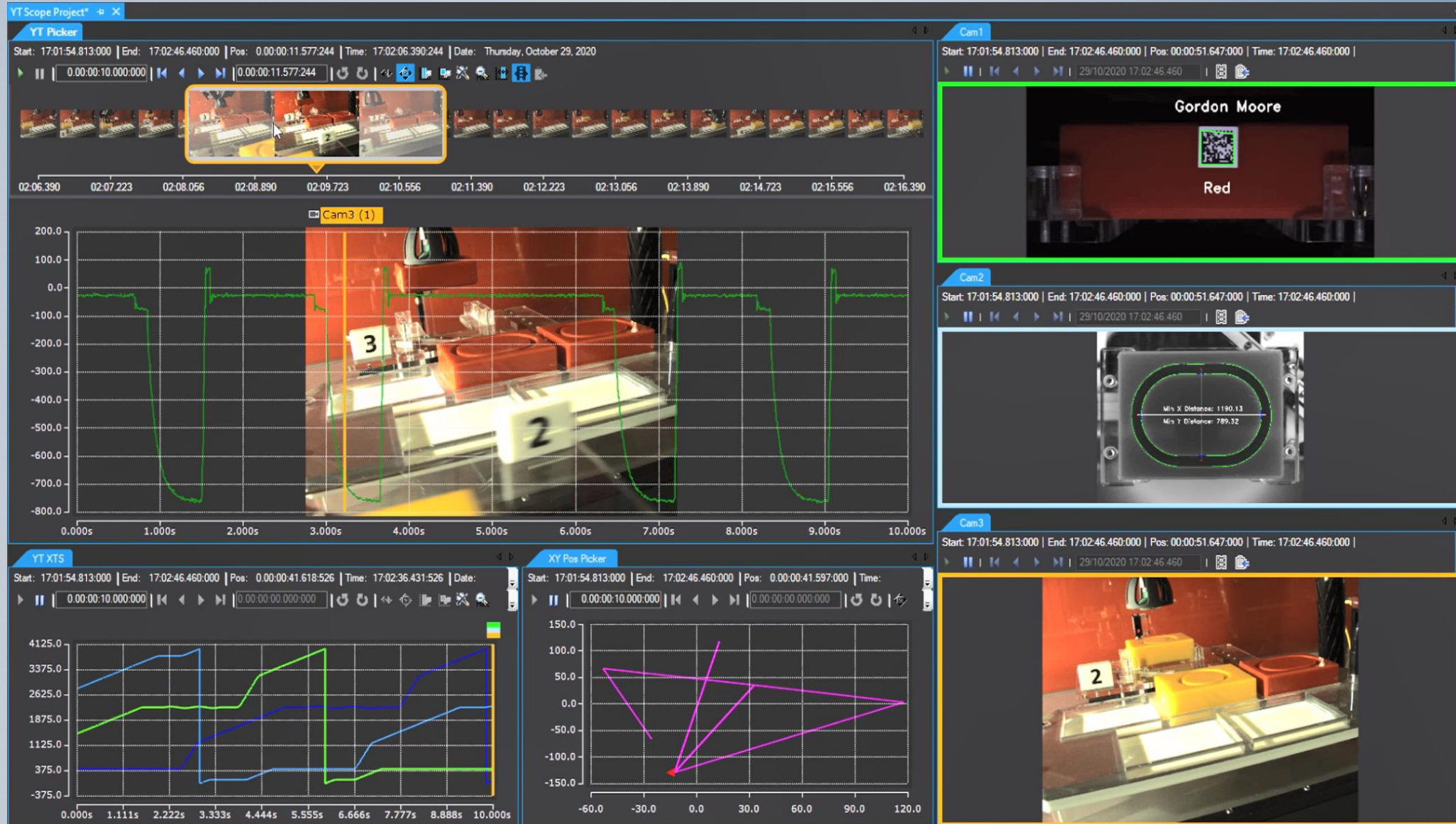


## Vision Integration

- Overlay image data (ITcVnDisplayableImage) with process data
- As single image or video stream
- Image acquisition settings:
  - Storage location, frame rate, compression
- A project can contain any number of images
  - a chart can contain only one image
- Performance implications
  - Use local Scope Server
  - Router memory should be increased
  - Reduce image resolution to reduced load
  - Lower FPS







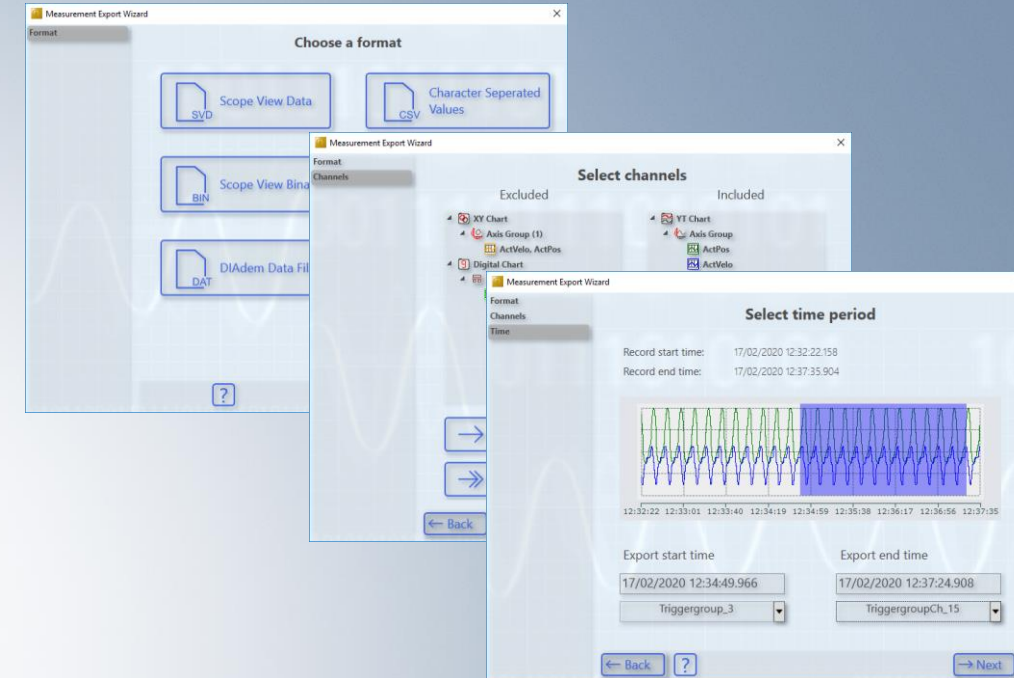
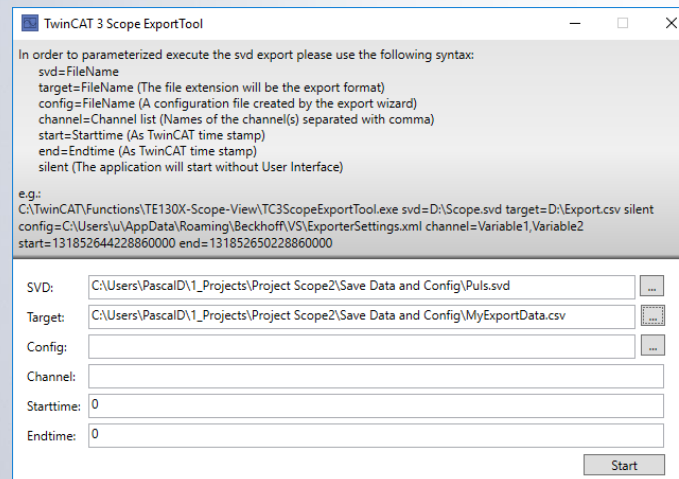
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# Data Handling and Exports

- Recordings can be exported from Scope View
  - Using Export Wizard
- Scope Server can record data without Scope View
  - Can be triggered by PLC
  - Local or remote systems





Scope Reporting

- Reports can be generated on-demand or triggered automatically

<< TwinCAT > Functions > TF3300-Scope-Server > TF3xxx-Reporting-Server > Reports > PDF

Name

Änderungsdatum

Typ

Größe

Report\_Beckhoff Report Template\_20220324T180905

24.03.2022 18:09

Adobe Acrobat-D...

118 KB

Report\_Beckhoff Report Template\_20220324T180525

Report\_Beckhoff Report Template\_20220324T173451

Report\_Beckhoff Report Template\_20220324T101616

Report\_Beckhoff Report Template\_20220322T153852

Report\_Beckhoff Report Template\_20220322T131731

Solution Explorer

TwinCAT\_Reporting\_Scope\_Template - Microsoft Visual Studio (Administrator)

File Edit View Project Build Debug TwinCAT TwinCAT HWB TwinSAFE PLC Team Scope Tools Test Analyze Window Help

Build 4024.17 (Loaded)

Solution: TwinCAT\_Reporting\_Scope\_Template (1 project)

TwinCAT\_Reporting\_Scope\_Template

VT Scope Project

DataPlot

Schalldruckpegel

Schwingungen

Axis Group (1)

Fine (1)

Marker (1)

Position

Axis Group (2)

Xi Fine

Yi FStairs (2)

Trigger

VT Scope Project

Schalldruckpegel

Schwingungen

Position

Time Marker (0)

Start: 09:46:42.375:700 End: 09:46:52.775:500 Pos: 0:00:00:00.400:800 Time: 09:46:42.775:500 Date: Donnerstag, 24. März 2022

0:00:00:10.000:000

20,0

16,0

12,0

8,0

4,0

0,0

1,000s

2,000s

3,000s

4,000s

5,000s

6,000s

7,000s

8,000s

9,000s

10,000s

Scope

Release Time

Comment

Scope

09:46:42.375:800

etc Reporting Trigs:

09:46:42.977:800

Scope

09:46:48.305:400

Scope

09:46:48.639:000

etc Reporting Trigs:

09:46:48.639:000

VT Chart (1) - Table (Referenced Chart: Schwingungen)

Select Marker

Position

Fine (1)

FStairs (1)

Marker (1)

10,7134

Time Marker (0)

04.023477 s

-11,95

10

Target Browser

Error List

Output

Find Symbol Results

Scope Reporting

Manual Report

Manual Scope Reporting

Report Name

Beckhoff Report Template

Title

DataKey

Include

Schalldruckpegel

Schalldruckpegel

☐

Schwingungen

Schwingungen

☐

Position

Position

☐

Chart Background

Light

Include Trigger Window

☐

Include Marker Window

☐

Generate Report

☐

Create

Quality management



Inspection according to manufacturer's specification

2022-05-02 12:17:08

General test information

|               |                                   |
|---------------|-----------------------------------|
| Manufacturer  | Beckhoff Automation GmbH & Co. KG |
| Test bench    | D2-750 bench 4                    |
| Description   | Drilling Machine MS 251           |
| Type          | MS 251 C-BE, RM3                  |
| Serial number | 253-5553-9421                     |

Analytics Workbench Overview

Additional Information

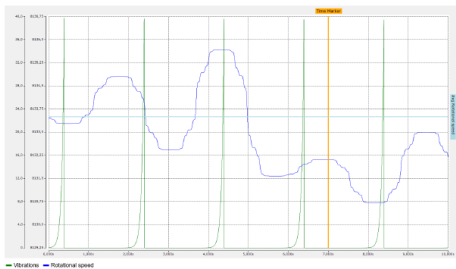
Operator: Max Mustermann

Measurement Information

Analysis Start Time: 02.05.2022 12:05:41

Analysis End Time: 02.05.2022 12:05:56

Rotations and Vibrations



Machine - Vibrations Min Max Avg

| Input | Min        | Max    | Avg    | Time Min                | Time Max                |
|-------|------------|--------|--------|-------------------------|-------------------------|
| 0     | 3,8147E-15 | 39,706 | 1,0054 | 02.05.2022 12:05:53.205 | 02.05.2022 12:05:43.202 |

Machine - Vibrations Overshoot

| Input | Operator Edge        | Threshold Edge | Edge  | Count | Last Event        |
|-------|----------------------|----------------|-------|-------|-------------------|
| 0     | GreaterThanOrEqualTo | 50             | False | 0     | No event occurred |



## Lab 2 – Scope Server

- Introduction, Licensing & Messaging
- Scope Basics
- Charts, Sampling & Triggers
- Layer Charts
- Dynamic Styles
- Vision Integration
- **Reporting and Data Export Formats**
- Additional Capabilities
- Tips & Tricks
- Q & A

[https://github.com/bstiffler582/2026\\_NEM\\_Scope](https://github.com/bstiffler582/2026_NEM_Scope)

Follow Lab 2 in **guide.md** file  
(best viewed in browser)



## Lab 3 – New Chart Types!

- Introduction, Licensing & Messaging
- Scope Basics
- Charts, Sampling & Triggers
- Layer Charts
- Dynamic Styles
- Vision Integration
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- Tips & Tricks
- Q & A

[https://github.com/bstiffler582/2026\\_NEM\\_Scope](https://github.com/bstiffler582/2026_NEM_Scope)

Follow Lab 3 in **guide.md** file  
(best viewed in browser)



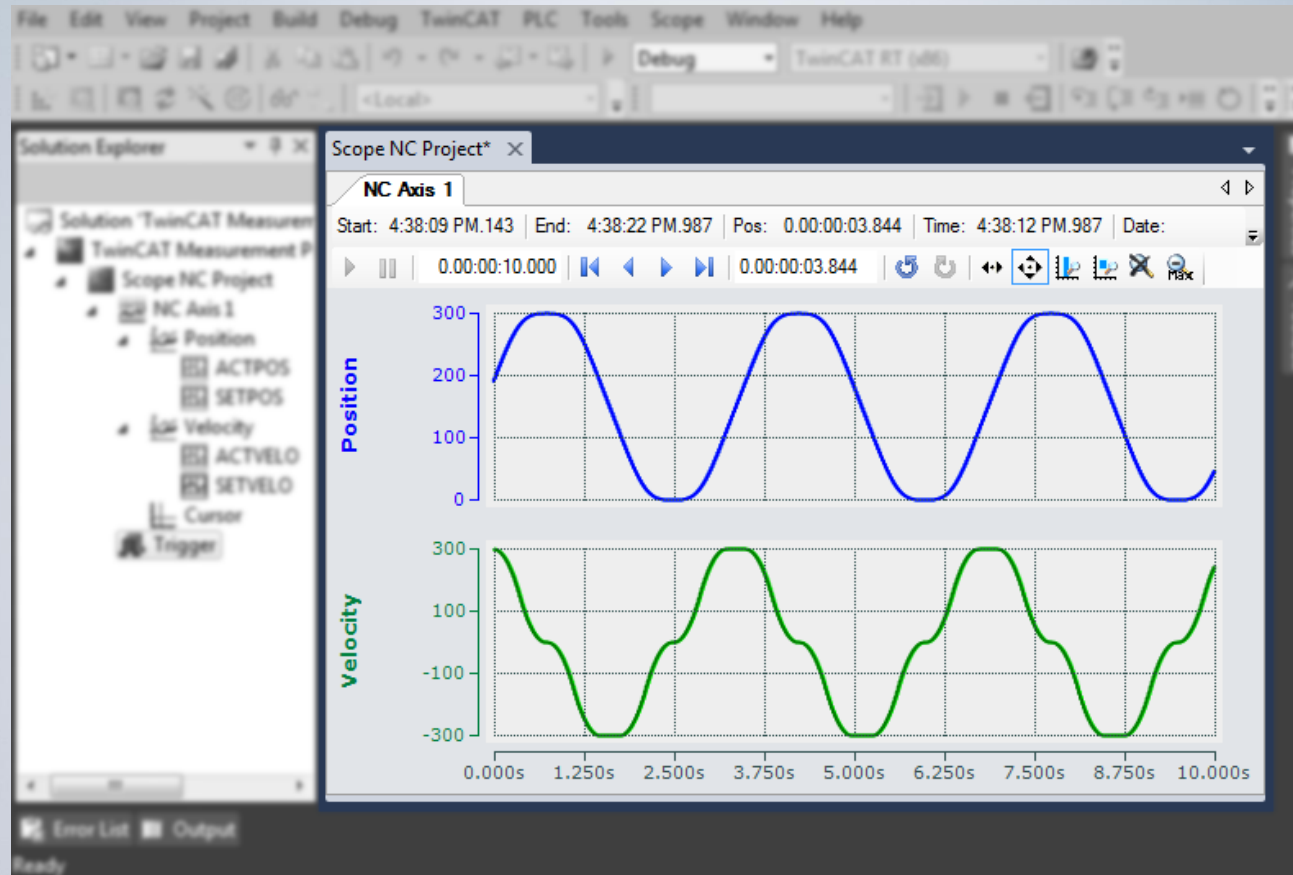
## NEM 2026 : Agenda

- Introduction, Licensing & Messaging
- Scope Basics
- Charts, Sampling & Triggers
- Layer Charts
- Dynamic Styles
- Vision Integration
- Reporting and Data Export Formats
- **Additional Capabilities**
- **Tips & Tricks**
- Q & A



## Scope Control Integration

- Create your own .NET based visualization and integrate a powerful and ready charting tool: TC3 Scope View Professional



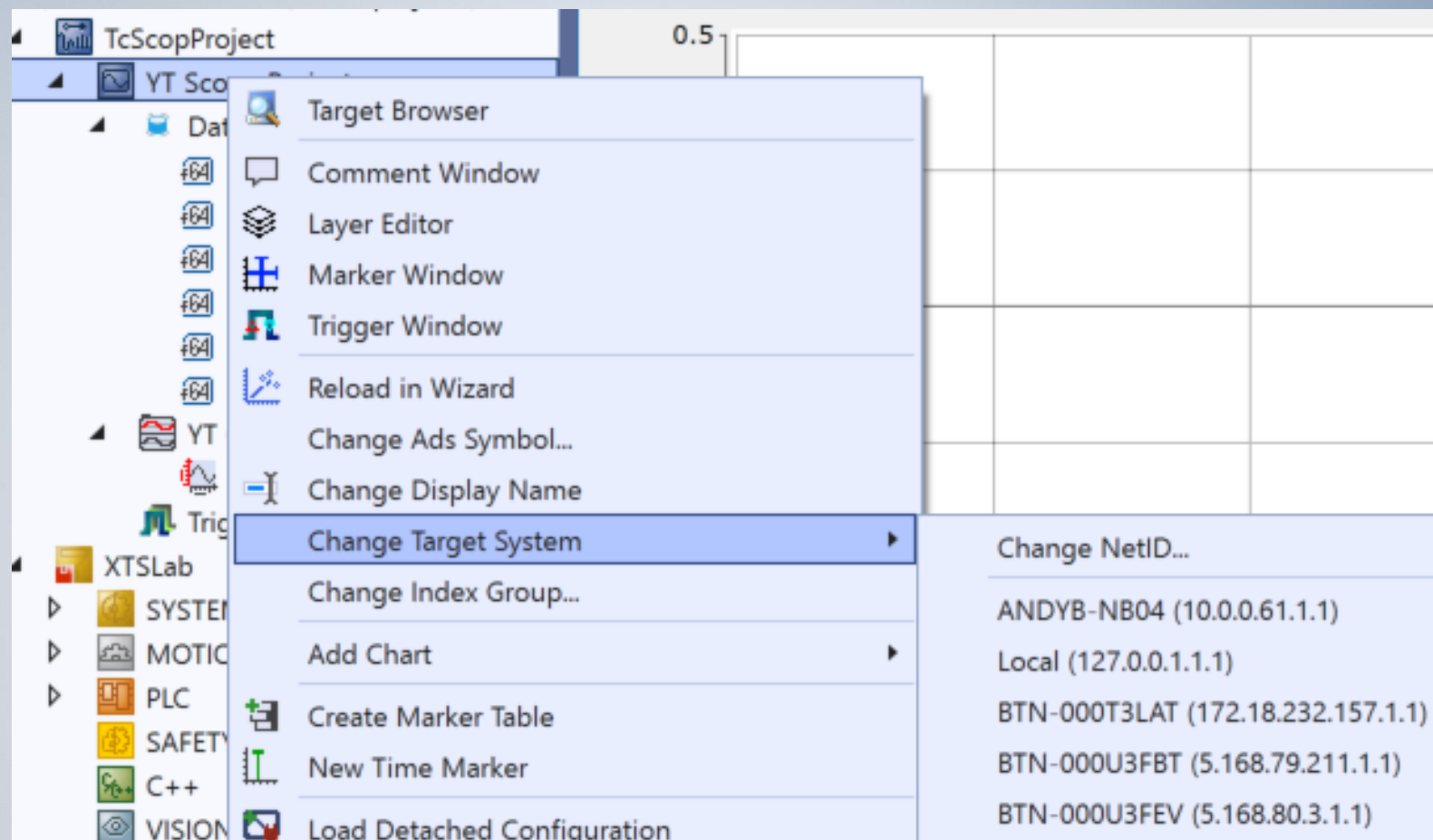
See Beckhoff Information System:

[http://infosys.beckhoff.com/content/1033/te13xx\\_tc3\\_scopeview/9007199437046923.html?id=16071](http://infosys.beckhoff.com/content/1033/te13xx_tc3_scopeview/9007199437046923.html?id=16071)



## Tips &amp; Tricks : Change Target System

BECKHOFF



# Tips & Tricks : Multiselect for Property Changes

Solution Explorer

Search Solution Explorer (Ctrl+;)

Solution 'XTSLab' (2 of 2 projects)

- TcScopProject
  - YT Scope Project
    - DataPool
      - ActPos
      - ActPosError
      - ActVelo
      - SetCurr
      - SetPosltp
      - SetVeloltp
    - YT Chart
      - Axis Group
      - Trigger

- XTSLab
- SYSTEM
- MOTION
- PLC
- SAFETY
- C++
- VISION
- ANALYTICS

Properties

Common

Symbol Comment

Enabled

Name

True

Symbol

Area

Data Type

Sample Mode

Sample Time [ms]

Time Offset[s]

Variable Size

Local

REAL64

Task Sampletime

0.25

0

8

Symbol ADS

Index Group

Index Offset

Symbol based

Symbol Name

0x1010220

True

Target

Target Port

Target System

Use Local Server

350

ANDYB-NB04 (10.0.0.61.1.1)

True

Unit

Unit

(None)

Solution Explorer

Search Solution Explorer (Ctrl+u)

Solution 'MyMeasurementProj' (2 projects)

- MyMeasurementProj
  - TwinCAT Measurement Project1
    - YT Scope Project
      - DataPool
      - YT Chart
        - Axis Group
          - fStairs
          - fPeak
          - fSine
        - Trigger

Properties

Common

Enabled

Name

Time Shift [µs]

Visible

True

0

True

Line

Antialias

Fill Color

Fill Mode

FillTransparency

Graph Type

Line Color

Line Width

True

None

50

Line

1

Marks

Mark Color

Mark Size

Mark State

2

Auto

Min/Max

Show Max

Show Min

False

False

Y-Data

Acquisition (Y)

Bitmask (Y)

Offset (Y)

Scalefactor (Y)

Unit (Y)

0xFFFFFFFFFFFFFFFF

0

1

(None)

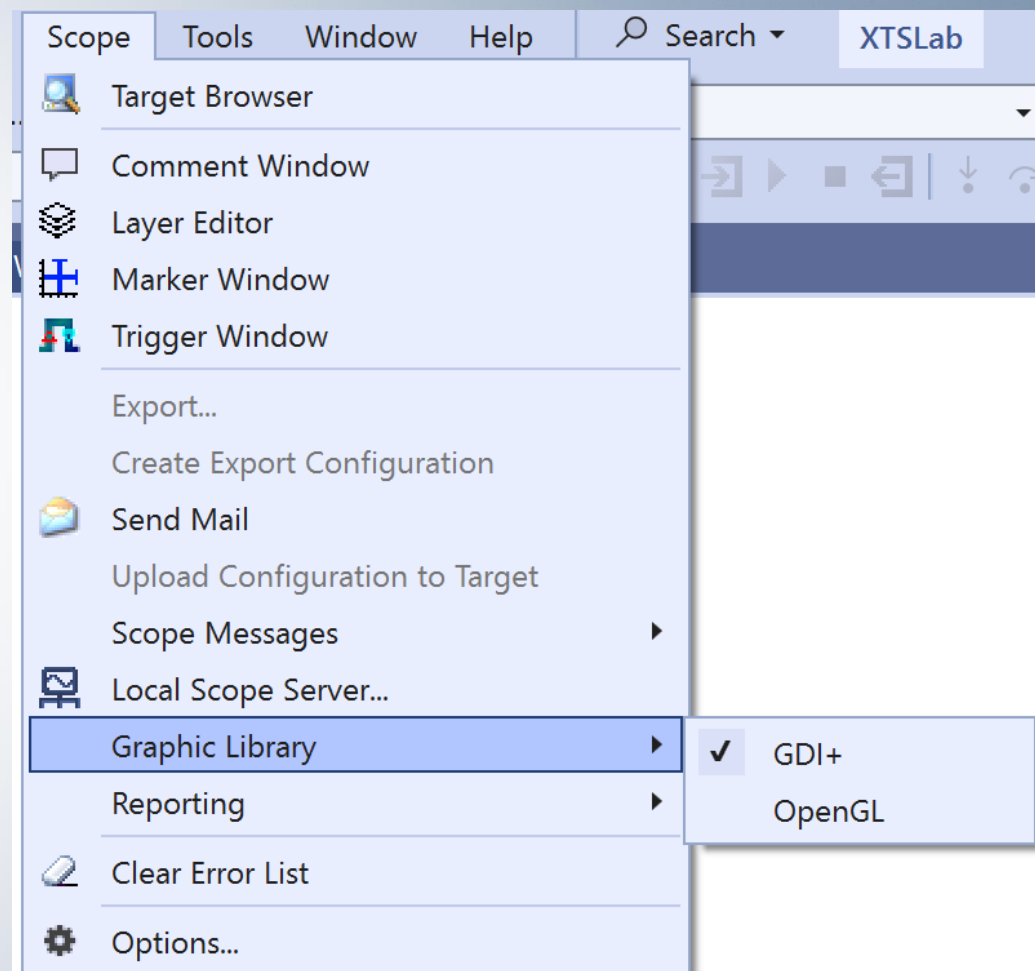
## Tips & Tricks : Graphics Driver

The graphics driver can be modified to optimize performance.

The default setting is GDI+, and is globally supported.

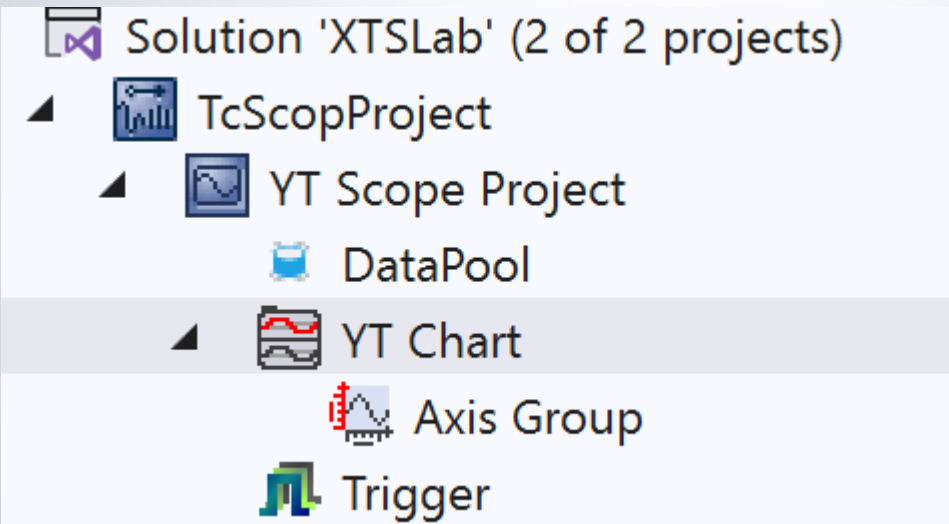
OpenGL optimizes rendering, which is required by some chart types.

OpenGL must be installed to the system.



# Tips & Tricks : CPU Core Selection

Charts can be associated to cores  
This can optimize performance.



Properties ▼ 📌 ✕

YT Chart YTChart - Properties ▼



Behaviour

|                        |                    |
|------------------------|--------------------|
| Auto Start             | True               |
| Data Tool Tip          | True               |
| Default Display Width  | 0.00:00:10.000:000 |
| Display overwrite Mode | False              |
| Master Chart           |                    |
| Time Bar               | True               |
| Tool Bar               | True               |

Color

|                     |                                                                                                   |
|---------------------|---------------------------------------------------------------------------------------------------|
| Border Color        |  240, 240, 240 |
| Forecolor           |  0, 0, 0       |
| Gradient Background | 1                                                                                                 |

Common

|           |                        |
|-----------|------------------------|
| Comment   |                        |
| CPU Core  | Default <span>▼</span> |
| Name      | YT Chart               |
| Show Name | False                  |

Overview

|            |                                                                                                     |
|------------|-----------------------------------------------------------------------------------------------------|
| Back Color |  255, 255, 255 |
| Fore Color |  0, 0, 0       |

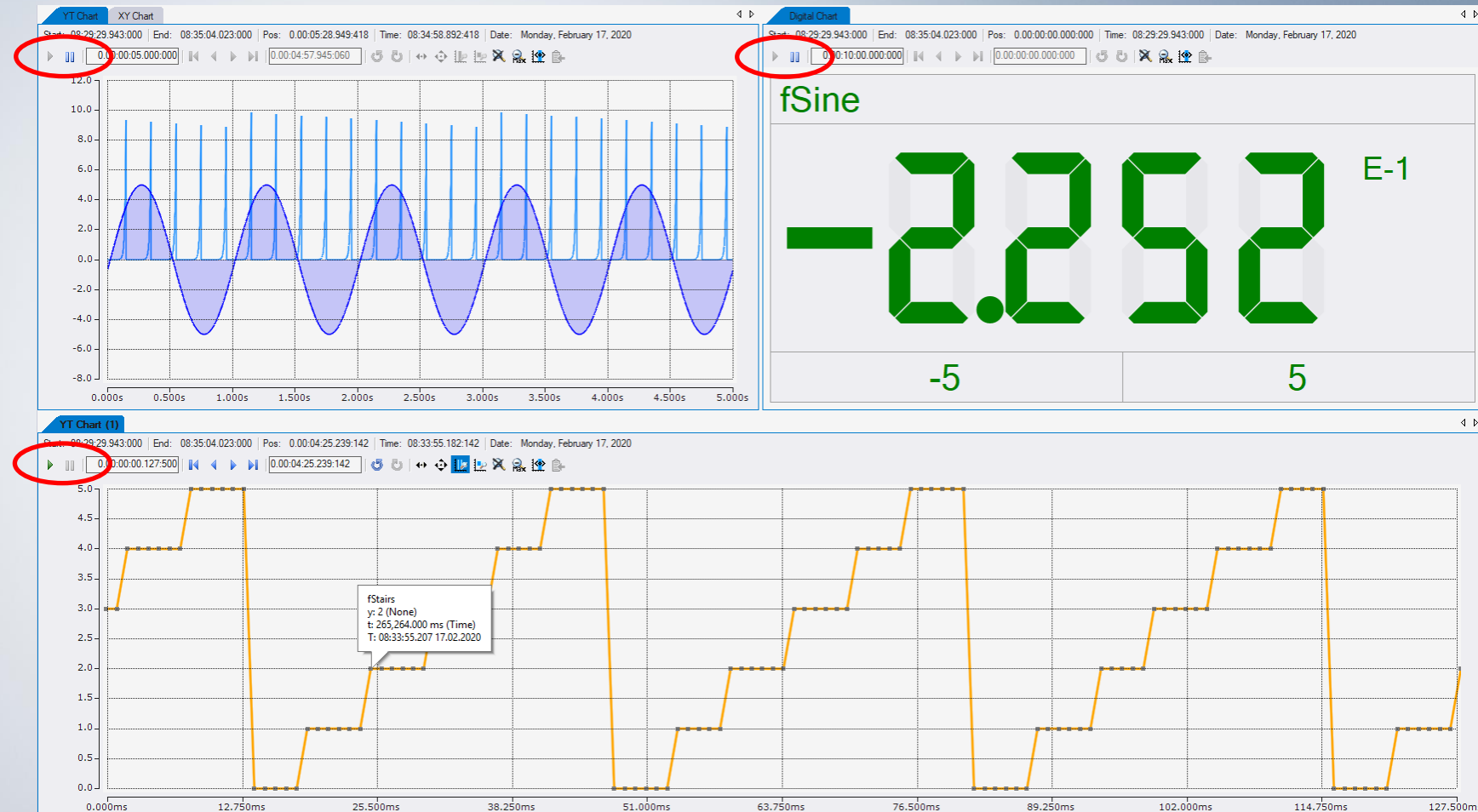
Settings

|               |       |
|---------------|-------|
| Scale On Zoom | True  |
| Stacked Axes  | False |



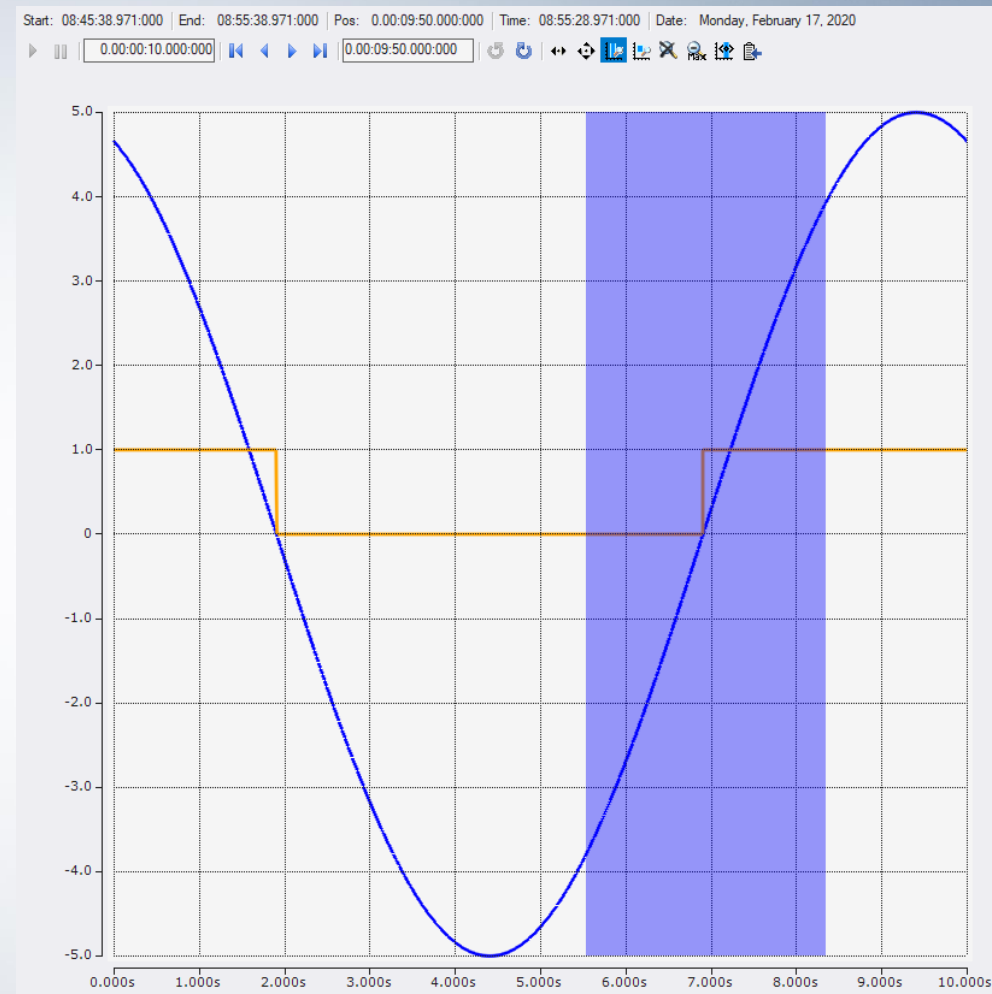
## Useful feature: Start/Stop Display

- Analysis during the record with Start/Stop Display
  - Recording start/stop != Display start/stop



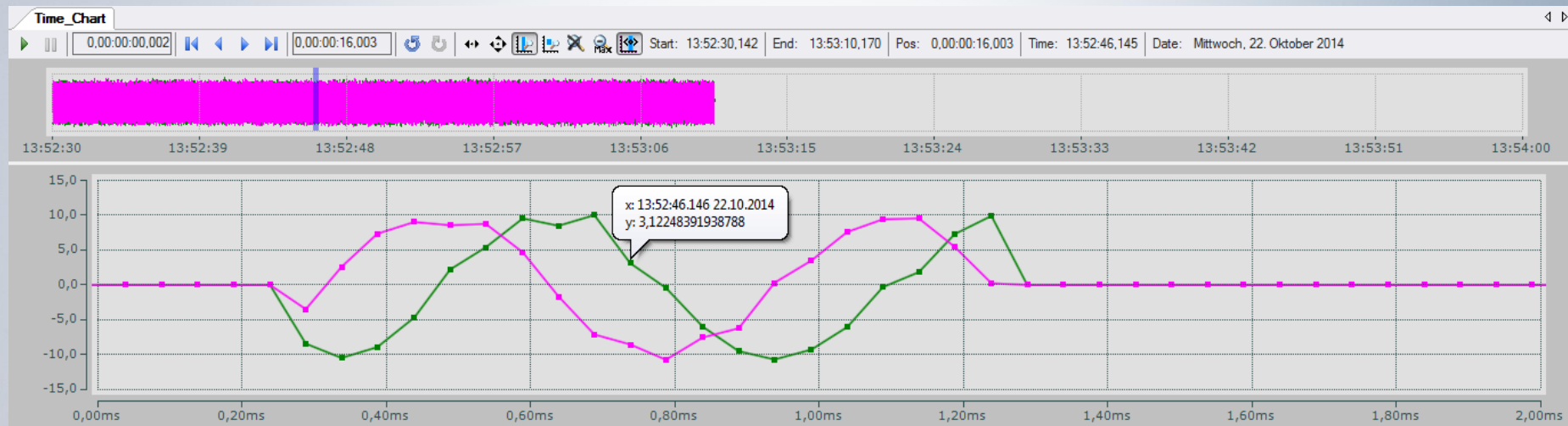
## Useful feature: Zoom and Panning

- Zoom X-Axis
- Zoom Y-Axis
- Zoom to Default
- Zoom Max Out
- Panning X Direction
- Panning X and Y Direction



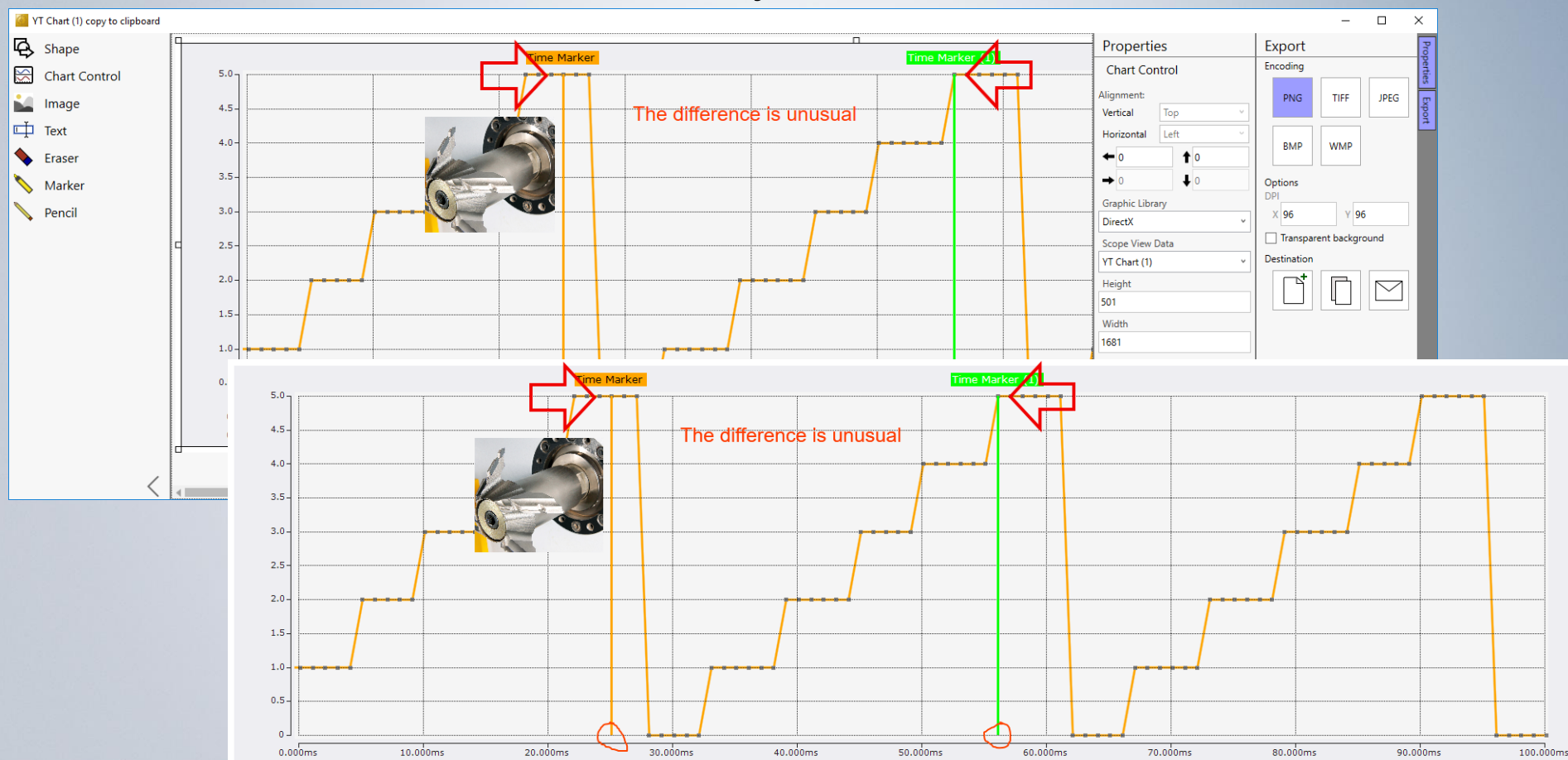
## Useful feature: Overview Chart

- Overview Chart in Chart with absolute time
- Marks
- Data picker



## Useful feature: Snipping Tool

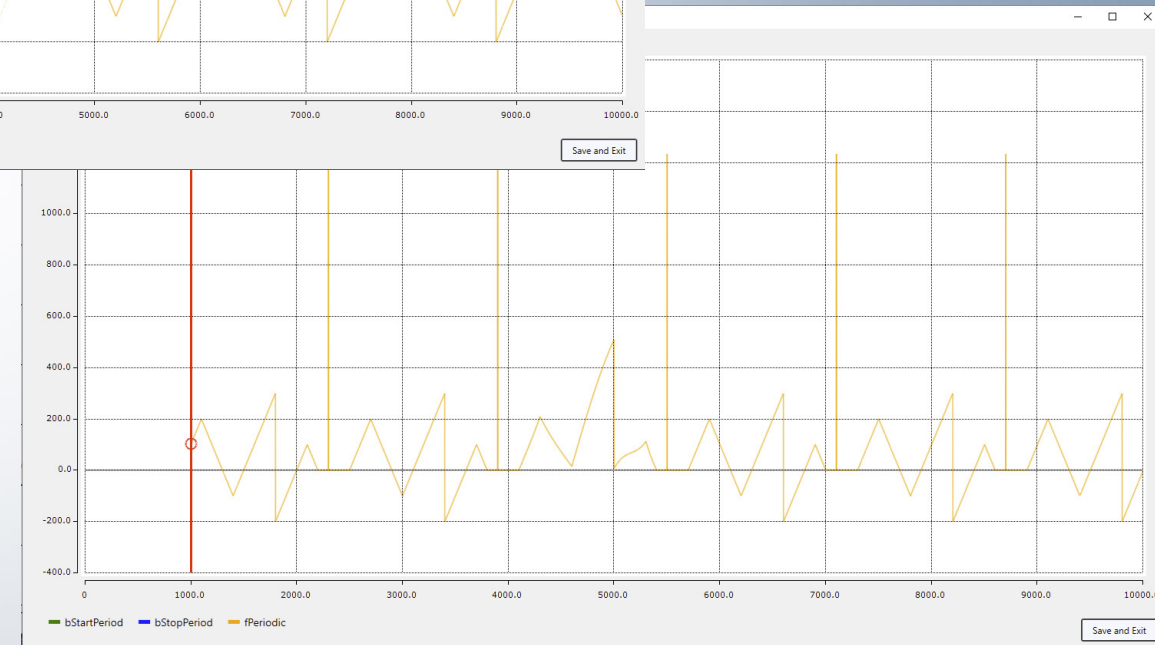
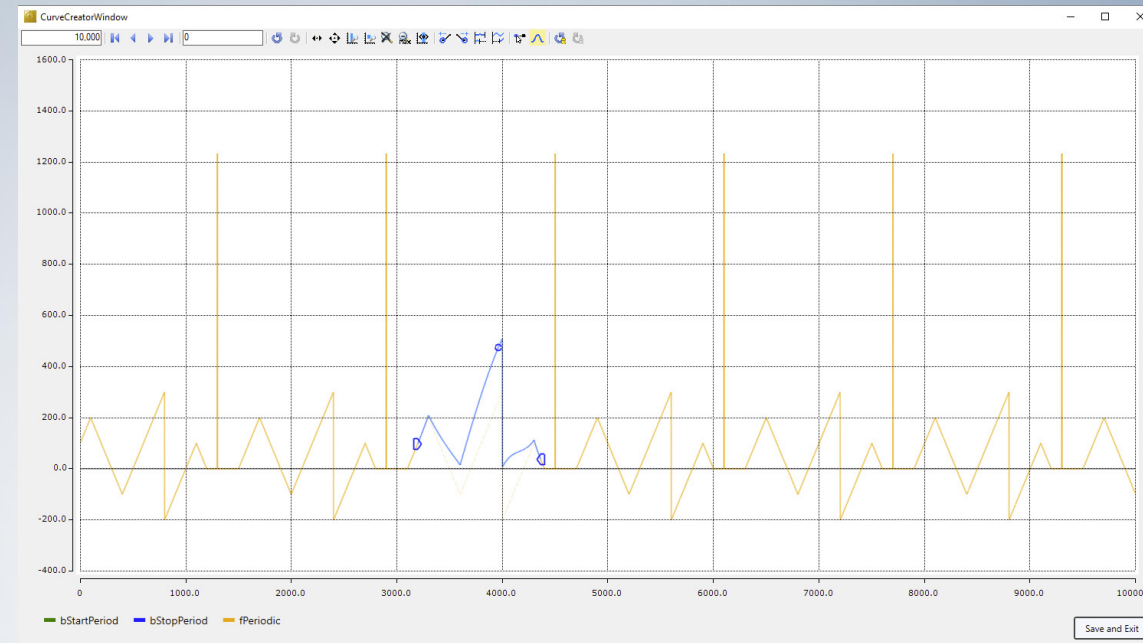
- Helps for documentation and sharing with colleagues
- Add shapes, text, pictures and mark your hints with a pencil
- Export in different formats, save or send by email





# TwinCAT Scope Curve Creator

- Design new channels
- Take snapshots
- Manipulate the curves
- Set start offset
- Stretch curves
- Squeeze curves
- Define start and end point
- Store the data as Offline Acquisition



## NEM 2026 : Agenda

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- **Q & A / Thanks!**

